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Categorical Semantics and Autonomous Theorem Proving

Autor: Carlos Fernández Lorán

Director: Dr. Carles Casacuberta and Dr. Joost J. Joosten

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Abstract

The main goal of this work is to prove the existence of.....

Notation: We will work over an algebraically closed field K of characteristic zero.

Introduction

The idea of generating mathematical reasoning computationally is one of the most profound challenges in computer science. At the core of this pursuit lies the Curry-Howard correspondence, an isomorphism between logic and computation [8]. Under this paradigm, mathematical propositions are interpreted as types, and the proofs of propositions are viewed as terms (or programs) inhabiting those types. Consequently, the problem of proving a theorem is reduced to the problem of finding a well-typed term in a formal type-theory.

Modern proof assistants, such as Rocq (formerly Coq) [5] and Lean [6], leverage this correspondence using rich type theories like the Calculus of Inductive Constructions [10]. While these environments enabled us to formalize complex theories, fully autonomous theorem proving with such rich frameworks remains an incredibly difficult challenge. The primary obstacle is undecidability of type inhabitation in dependent type theories: attempting to exhaustively construct a term of a given type results in a combinatorial explosion of the search space.

Traditionally, term generation has been approached from a purely syntactic perspective. This has more recently been the area of work of some teams using Large Language Models. But generating terms just from the syntax is very difficult to guide heuristically. Our work trying to understand the framework proposed by Laurent Lafforgue that connects Topos Theory [9] with artificial intelligence, lead to the following redefinition of the problem: leveraging the categorical semantics of type theories to reframe autonomous theorem proving as a structural graph exploration problem. By translating the syntax of Generalized Algebraic Theories (GATs) and the Calculus of Constructions (CoC) into categorical structures—namely Contextual Categories and Doctrines of Constructions—we obtain term models that inherently possess a rich, geometric graph structure. The soundness of this translation is guaranteed by the term model being the initial object of the category of models of the underlying type theory. This result is known as the *initiality conjecture* from Voevodsky [12]. It was first proven by Streicher in [11] for CoC, and more recently for a full-featured Martin-Löf type theory in September 2020 by Brunerie and Lumsdaine in [2].

Withing this categorical framework, the problem of finding a proof translates to finding a specific section (a morphism) withing a relativized contextual category. With these idea in mind of using categories and neural networks, we formalized an expansion algorithm—a deterministic process that exhaustively builds the finite subcategories representing the search space of terms—specifically designed to work together with Graph Neural Networks (GNNs), and try to overcome these intrinsic severe exponential growth inherent to any algorithm trying to exhaustivity solve the problem.

Graph Neural Networks are used to navigate and prune the categorical search space. Considering our finite subcategories that form the known search space as graphs, they are the perfect inputs for message-passing neural networks. By training a GNN to classify nodes in the context graph (deciding witch paths to expand, keep or delete), we can effectively guide the expansion algorithm, bypassing the combinatorial explosion and paving the way for efficient, autonomous theorem proving.

Objectives

In this work we first aim to provide a proof of the initiality conjecture for GATs, together with an extension of Streicher’s proof for CoC together with a GAT signature. Then, once established the mathematical significance of objects and morphisms in the term model, we then present the expansion algorithm, an algorithm to construct any object or morphism in a Contextual Category or Contextual Doctrine, and that is designed to work together with a graph neural network to efficiently solve the problem in hand. Then we implement the expansion algorithm for simply typed combinatory logic, as a simple example to give intuition about how the expansion process works. Finally we give a proposal of a GNN-based model thought to optimize term generation.

Thesis Outline

This thesis is structured to guide the reader from the abstract categorical foundations to the practical implementation of AI-guided theorem proving:

- **Chapter 1: Syntax and Semantics of Generalized Algebraic Theories.** We introduce Cartmell’s Generalized Algebraic Theories and define their syntax and semantics, as they were originally presented in [3] [4] and incorporating some notions presented by Streicher in [11]. Then we provide a concept of model of a GAT U together with a proof of the initiality of the term model

$C(U)$. Finally, define the problem of type inhabitation in categorical terms, and present the foundational Expansion Algorithm.

- **Chapter 2: Calculus of Constructions.** We step up the complexity to the Calculus of Constructions (CoC), introducing product types and the universe of propositions. We present the definition of CoC together with the construction of Doctrines of Constructions as models for CoC, given by Streicher in [11]. We extend the definition of CoC to include a GAT signature, define the new term model and prove its initiality. Finally, we extend our categorical problem definition and Expansion Algorithm to handle this richer syntax.
- **Chapter 3: Implementation of the Expansion Algorithm.** We bring the theory into practice. We demonstrate the raw Expansion Algorithm generating terms of Simply Typed Combinatory Logic ($CL_{\rightarrow}$) using `IMPILICA`, a custom Rust-based graph engine. Finally, we outline the architecture for integrating Graph Neural Networks using a novel unsupervised "Conjecturer-Prover" training paradigm to optimize the expansion process.

Chapter 1

The Syntax and Semantics of Generalized Algebraic Theories

In this chapter, we introduce Generalized Algebraic Theories (GATs), a framework for working with dependent types. Our ultimate goal in this thesis is to translate the syntactical problem of theorem proving into a geometrical graph-exploration problem. GATs allow us to construct the concept of *contextual category*, the foundational categorical structure to deal with dependent types. We use this structure as a baseline for interpreting the more complex Calculus of Constructions in Chapter 2.

1.1 Generalized Algebraic Theories

Generalized Algebraic Theories, originally presented by J. Cartmell in his thesis [3] [4], is an abstraction of Martin-Löf type theory and is a generalization of the usual notion of a many-sorted algebraic theory in the following manner: while in Martin Löf’s many-sorted algebraic theories sorts are constant types, interpreted as sets, in generalized algebraic theories sorts might be constant or variable, over a specified range of types. Variable types are interpreted as families of sets.

Formally, a generalized algebraic theory consists of:

1. a set of sorts, that can be constant or variable;
2. a set of operators, with the sorts of the arguments and value specified;
3. a set of axioms, where each axiom is an identity between similar expressions.

To build intuition, consider the theory of categories. The set of sorts contains two elements *Ob* and *Hom*. *Ob* is a constant type and *Hom* is a variable type;

for any two terms x, y of type Ob , we have a type $\text{Hom}(x, y)$. Here, we see a key feature of GATs: types vary over terms, not over types. The set of operators also has two elements id and $\circ$. The id operator has one argument x of type Ob , and returns a value of type $\text{Hom}(x, x)$. This shows how the type of value of an operator might vary depending on the arguments. The $\circ$ operator (or composition) expects 5 arguments, three terms x, y, z of type Ob , one term f of type $\text{Hom}(x, y)$, and one term g of type $\text{Hom}(y, z)$. It returns a value of type $\text{Hom}(x, z)$. Here we see that the type of an argument might also vary depending on a previous argument. Informally, instead of writing $\circ(x, y, z, f, g)$ we can write $\circ(f, g)$, as the first three arguments might be inferred from the last two. This notation is discussed in the original paper [4].

Now we introduce the notation we will be using when talking about terms and types. This notation will be formally defined in a later section. Given a set of variables V , we associate terms to variables as required. When asserting that a term t is of type A , we will write $t : A$, but if t has variables $x_1, \dots, x_n$ occurring in it, this assertion does not make sense unless we assign some types to $x_1, \dots, x_n$. To do this we write $x_1 : A_1, \dots, x_n : A_n \vdash t : A$ or

$$\frac{x_1 : A_1, \dots, x_n : A_n}{t : A}$$

that reads as follows: if x_1 is a variable of type $A_1, \dots$ and if x_n is a variable of type A_n , then t is a term of type A . Similarly, if we want to assert that if x_1 is a variable of type $A_1, \dots$ and if x_n is a variable of type A_n , then A is a type, we can write: $x_1 : A_1, \dots, x_n : A_n \vdash A \text{ type}$ or

$$\frac{x_1 : A_1, \dots, x_n : A_n}{A \text{ type}}$$

Such assertions are called rules or sequents. Rules are used to express which expressions are well-formed. We work with rules as units instead of the base expressions. This is because terms and types do not make sense by themselves but they need to be thought of in a context.

Axioms are also written as rules. The conclusion of those rules are equalities between types or between terms of the same type. We write those rules as follows:

$$\frac{x_1 : A_1, \dots, x_n : A_n}{A = A'} \qquad \frac{x_1 : A_1, \dots, x_n : A_n}{t = t' : A}$$

and read as: if x_1 is a variable of type $A_1, \dots$ and x_n is a variable of type A_n , then $A = A'$, or $t = t'$ as terms of type A , respectively. We will be using both shorthand and fraction notation as needed.

To continue with the previous example, we can write the axioms of category

theory as follows:

$$\begin{aligned} x, y : Ob, f : Hom(x, y), &\vdash \circ(id(x), f) = f \\ x, y : Ob, f : Hom(x, y), &\vdash \circ(f, id(y)) = f \\ w, x, y, z : Ob, f : Hom(w, x), g : Hom(x, y), h : Hom(y, z) \\ &\vdash \circ(\circ(f, g), h) = \circ(f, \circ(g, h)) \end{aligned}$$

When presenting a theory's language i.e., the sets of sorts and operators, we must present each symbol with a rule specifying the role of this symbol, either as a sort symbol or as specifically typed operator symbol. These rules are called introductory rules. In the case of a sort symbol A , the rule should be like

$$\frac{x_1 : A_1, \dots, x_n : A_n}{A(x_1, \dots, x_n) \text{ type}}$$

clearly specifying the types A is dependent on. In the case of an operator symbol, a rule of the form

$$\frac{x_1 : A_1, \dots, x_n : A_n}{f(x_1, \dots, x_n) : A}$$

suffices to specify the types of the arguments and the value of f .

Finally, to present a theory we will give a set of symbols, each associated to its introductory rule, and a set of axioms. And, of course, everything must be well-formed, **something we will address in the following section**. So now we can give a formal presentation of the theory of categories:

Symbol	Introductory Rule
--------	-------------------

Ob	Ob is a type.
Hom	$x, y \in \text{Ob} : \text{Hom}(x, y)$ is a type.
$\circ$	$x, y, z \in \text{Ob}, f \in \text{Hom}(x, y), g \in \text{Hom}(y, z) : \circ(f, g) \in \text{Hom}(x, z).$
id	$x \in \text{Ob} : \text{id}(x) \in \text{Hom}(x, x).$

Axioms

$$\begin{aligned} \circ(\text{id}(x), f) &= f, & \text{whenever } x, y \in \text{Ob} \text{ and } f \in \text{Hom}(x, y). \\ \circ(f, \text{id}(y)) &= f, & \text{whenever } x, y \in \text{Ob} \text{ and } f \in \text{Hom}(x, y). \\ \circ(\circ(f, g), h) &= \circ(f, \circ(g, h)), & \text{whenever } w, x, y, z \in \text{Ob}, f \in \text{Hom}(w, x), \\ & & g \in \text{Hom}(x, y) \text{ and } h \in \text{Hom}(y, z). \end{aligned}$$

1.2 Formal Definition

In this section we formally define all of the concepts we just introduced. To give a notion of well-formed introductory rule, we need the concept of derivability,

which also depends on the introductory rules. To tackle this problem, we leave aside the question of well-formedness until we have the complete set of derived rules of a theory. We first define a pretheory and then define a theory as a well-formed pretheory.

We assume given a countable set of variables V . Let us define the notion of rule on an alphabet W , whose elements are called symbols.

Definition 1.1. The set of *expressions* $\mathcal{E}$, is defined inductively so that expressions are finite sequences of elements of $W \cup V \cup \{(\} \cup \{)\} \cup \{, \}$ defined by the following clauses:

1. if $x \in V$, then x is an expression;
2. if $L \in W$, then L is an expression;
3. if $L \in W$ and $e_1, \dots, e_n$ are expressions, then $L(e_1, \dots, e_n)$ is an expression.

Now we can define the building blocks of a rule:

Definition 1.2. A *premise* is defined to be a sequence of $V \times \mathcal{E}$. The empty sequence is allowed and we call it an empty premise. If $((x_1, A_1), \dots, (x_n, A_n))$ is a premise, we write it as $x_1 : A_1, \dots, x_n : A_n$.

Definition 1.3. We call a *conclusion* to one of the following:

1. T -conclusion: defined by a single expression A and written as ' A is a type'.
2. $\in$ -conclusion: defined by a pair of expressions (t, A) and written as ' $t : A$ '.
3. $T =$ conclusion: defined by a pair of expressions (A_1, A_2) and written as ' $A_1 = A_2$ '.
4. $\in =$ conclusion: defined by a triple of expressions (t_1, t_2, A) and written as ' $t_1 = t_2 : A$ '.

Finally a *rule* is determined by a premise P and a conclusion C , and is written as $\frac{P}{C}$ or as $P \vdash C$. We name the rule as T -rule, $\in$ -rule, $T =$ rule or $\in =$ rule according to the form of the conclusion.

Now we can define a pretheory and later a theory.

Definition 1.4. A **pretheory** consists of

1. a set of sort symbols S ;
2. a set of operator symbols Z ;

3. for each sort symbol A , an associated rule of the alphabet $S \cup Z$:

$$\frac{x_1 : A_1, \dots, x_n : A_n}{A(x_1, \dots, A_n) \text{ type}}$$

for some $n \geq 0$. It is called the *introductory rule* of A ;

4. for each operator symbol F , an associated rule of the alphabet $S \cup Z$:

$$\frac{x_1 : A_1, \dots, x_n : A_n}{F(x_1, \dots, x_n) : A}$$

for some $n \geq 0$. It is called the *introductory rule* of F ;

5. a set of axioms. Where each axiom is either a $T =$ rule or a $\in =$ rule of the alphabet $S \cup Z$.

Now we tackle the problem of well-formedness of rules.

Definition 1.5. If U is a pretheory, then

1. a **context** is a premise $x_1 : \Delta_1, \dots, x_n : \Delta_n$ such that the rule

$$\frac{x_1 : \Delta_1, \dots, x_{n-1} : \Delta_{n-1}}{\Delta_n \text{ is a type}}$$

is a derived rule such that x_n is distinct from all of $x_1, \dots, x_{n-1}$.

2. (a) the rule $\frac{x_1 : \Delta_1, \dots, x_n : \Delta_n}{\Delta \text{ is a type}}$ is **well-formed** if the premise $x_1 : \Delta_1, \dots, x_n : \Delta_n$ is a context.
- (b) the rule $\frac{x_1 : \Delta_1, \dots, x_n : \Delta_n}{t : \Delta}$ is **well-formed** if $\frac{x_1 : \Delta_1, \dots, x_n : \Delta_n}{\Delta \text{ is a type}}$ is a derived rule of U .
- (c) the rule $\frac{x_1 : \Delta_1, \dots, x_n : \Delta_n}{\Delta = \Delta'}$ is **well-formed** if $\frac{x_1 : \Delta_1, \dots, x_n : \Delta_n}{\Delta \text{ is a type}}$ and $\frac{x_1 : \Delta_1, \dots, x_n : \Delta_n}{\Delta' \text{ is a type}}$ are derived rules of U .
- (d) the rule $\frac{x_1 : \Delta_1, \dots, x_n : \Delta_n}{t = t' : \Delta}$ is **well-formed** if $\frac{x_1 : \Delta_1, \dots, x_n : \Delta_n}{t : \Delta}$ and $\frac{x_1 : \Delta_1, \dots, x_n : \Delta_n}{t' : \Delta}$ are derived rules of U .

Definition 1.6. The set of *derived rules* of a pretheory U is the set of rules derivable by the derivation rules presented in Appendix A.1.

Definition 1.7. A pretheory is *well-formed* if all of its introductory rules are well-formed. A **theory** is a well-formed pretheory.

Lastly, we introduce substitution. Given expressions $\Delta, t_1, \dots, t_n$ and variables $x_1, \dots, x_n$, then we define the expression $\Delta[x_1|t_1, \dots, x_n|t_n]$ as the result of simultaneous replacement of $x_1, \dots, x_n$ in Δ by $t_1, \dots, t_n$. We call this the Substitution operation. Then, as proven in the original paper [4], the set of derived rules of a theory is closed under the operation of substitution of correctly typed terms for variables. We could add substitution as one of the derivation rules, but this would make applying induction very messy.

1.3 Contexts and Realizations

To transition from syntax to geometry (which our Graph Neural Network will eventually navigate), we construct a category out of the theory syntax.

For a theory U , we define the category of contexts and realizations, denoted $R(U)$. This category has as objects the contexts of U . Given two contexts $\Gamma \equiv x_1 : A_1, \dots, x_n : A_n$ and $\Delta \equiv y_1 : B_1, \dots, y_m : B_m$ of U , a morphism from Γ to Δ , or a *realization* of Δ from Γ , is an m -tuple $\langle t_1, \dots, t_m \rangle$ such that for each j , $1 \leq j \leq m$, the rule

$$\Gamma \vdash t_j : B_j[y_1 \mid t_1, \dots, y_{j-1} \mid t_{j-1}]$$

is derivable. The identity of an object $x_1 : \Delta_1, \dots, x_n : \Delta_n$ is the realization $\langle x_1, \dots, x_n \rangle$, and composition is defined as

$$\begin{aligned} \langle t_1, \dots, t_m \rangle \circ \langle s_1, \dots, s_w \rangle = \\ \langle s_1[y_1 \mid t_1, \dots, y_m \mid t_m], \dots, s_w[y_1 \mid t_1, \dots, y_m \mid t_m] \rangle \end{aligned}$$

whenever $\Gamma = x_1 : A_1, \dots, x_n : A_n$, $\Delta = y_1 : B_1, \dots, y_m : B_m$ and $\Omega = z_1 : C_1, \dots, z_w : C_w$ are contexts, and $\langle t_1, \dots, t_m \rangle : \Gamma \rightarrow \Delta$ and $\langle s_1, \dots, s_w \rangle : \Delta \rightarrow \Omega$ are realizations.

Now we study some properties of this category. First, we see that it has a tree structure.

Definition 1.8. A tree is a pair (N, father) where N is a nonempty set of nodes and $\text{father} : N \rightarrow N$ is a function such that:

1. for all $A \in N$, there exists $n \in \omega$ such that $\text{father}^n(A) = \text{father}^{n+1}(A)$;
2. for all $A, B \in N$, if $A = \text{father}(A)$ and $B = \text{father}(B)$, then $A = B$.

We write $A \sqsubseteq B$ if $A = \text{father}(B)$ and $A \triangleleft B$ if $A \sqsubseteq B$ and $A \neq B$. By condition 2, there is a unique least element of N , which we call 1. Finally, we will define, for any $A \in N$, $\text{level}(A)$ to be the least $n \in \omega$ with $\text{father}^n(A) = \text{father}^{n+1}(A)$. We can see that the set of objects of $R(U)$, i.e., the set of contexts of U , is structured as a tree with the empty context as its least element

and $\text{father}(\langle x_1 : A_1, \dots, x_n : A_n \rangle) = \langle x_1 : A_1, \dots, A_{n-1} \rangle$. Moreover, given a context $x_1 : A_1, \dots, x_n : A_n, x_{n+1} : A_{n+1}, \dots, x_{n+m} : A_{n+m}$ of U , $\langle x_1, \dots, x_n \rangle$ is a realization of $x_1 : A_1, \dots, x_n : A_n$ with regard to $x_1 : A_1, \dots, x_n : A_n, x_{n+1} : A_{n+1}, \dots, x_{n+m} : A_{n+m}$, and is noted as $p(\langle x_1 : A_1, \dots, x_{n+m} : A_{n+m} \rangle, \langle x_1 : A_1, \dots, x_n : A_n \rangle)$. Hence, given two objects $\Gamma, \Delta \in \text{Ob}R(U)$ such that $\Gamma \leq \Delta$, $p(\Delta, \Gamma)$ is defined and $p(\Delta, \Gamma) : \Delta \rightarrow \Gamma$ in $R(U)$.

If $\Gamma \equiv x_1 : A_1, \dots, x_n : A_n, \Gamma' \equiv y_1 : B_1, \dots, y_m : B_m$ and $\Delta \equiv y_1 : B_1, \dots, y_{m+w} : B_{m+w}$ are contexts, and if $f = \langle t_1, \dots, t_m \rangle : \Gamma \rightarrow \Gamma'$ in $R(U)$, then

$$\begin{array}{ccc} f^* \Delta & \xrightarrow{q(f, \Delta)} & \Delta \\ p(f^* \Delta, \Gamma) \downarrow & & \downarrow p(\Delta, \Gamma') \\ \Gamma & \xrightarrow{f} & \Gamma' \end{array}$$

is a pullback diagram in $R(U)$, where

$$\begin{aligned} f^* \Delta &\equiv x_1 : A_1, \dots, x_n : A_n, \\ y_{m+1} : B[y_1 \mid t_1, \dots, y_m \mid t_m], \dots, y_{m+w} : B[y_1 \mid t_1, \dots, y_m \mid t_m] \\ q(f, \Delta) &= \langle t_1, \dots, t_m, y_{m+1}, \dots, y_{m+w} \rangle. \end{aligned}$$

This motivates the following definition.

1.4 Contextual Categories

Definition 1.9. A contextual category consists of

1. a category $\mathcal{C}$ with a terminal object 1;
2. a function $\text{father} : \text{Ob}\mathcal{C} \rightarrow \text{Ob}\mathcal{C}$ such that $(\text{Ob}\mathcal{C}, \text{father})$ is a tree, and 1 is the root of the tree, i.e., $\text{father}1 = 1$;
3. for each object A of $\mathcal{C}$ different from 1, a morphism $p(A) : A \rightarrow \text{father}A$, called *canonical projection of A* , which we write as $A \twoheadrightarrow \text{father}(A)$;
4. for all objects A, B in $\mathcal{C}$ with $A \triangleleft B$ and all morphisms $f : C \rightarrow A$, an object f^*B of $\mathcal{C}$ such that $C \triangleleft f^*B$, and a morphism $q(f, B) : f^*B \rightarrow B$ in $\mathcal{C}$ such that

$$\begin{array}{ccc} f^*B & \xrightarrow{q(f, B)} & B \\ \downarrow & & \downarrow \\ C & \xrightarrow{f} & A \end{array}$$

is a pullback in $\mathcal{C}$ that satisfies the following *functoriality conditions*:

(a) if A and B are objects in $\mathcal{C}$ such that $A \triangleleft B$, then

$$id_A^* B = B \quad \text{and} \quad q(id_A, B) = id_B;$$

(b) whenever

$$\begin{array}{ccccc} & & & & B \\ & & & & \downarrow \\ A & \xrightarrow{f} & A' & \xrightarrow{f'} & A'' \end{array}$$

in $\mathcal{C}$, then

$$(f' \circ f)^* B = f^*(f'^* B) \quad \text{and} \quad q(f' \circ f, B) = q(f', B) \circ q(f, f'^* B).$$

We have that $R(U)$ is structured as a contextual category. The problem with $R(U)$ is that semantically equal but syntactically different rules are interpreted as different elements. Hence we define an equivalence relation between derived T - and $\in$ - rules as follows:

$$x_1 : A_1, \dots, x_n : A_n \vdash A_{n+1} \text{ type} \equiv y_1 : B_1, \dots, y_m : B_m \vdash B_{m+1} \text{ type}$$

if and only if $m = n$ and, for each $1 \leq i \leq n + 1$,

$$x_1 : A_1, \dots, x_{i-1} : A_{i-1} \vdash A_i = B_i[y_1 \mid x_1, \dots, y_{i-1} \mid x_{i-1}]$$

is derivable, and

$$x_1 : A_1, \dots, x_n : A_n \vdash t : A \equiv y_1 : B_1, \dots, y_m : B_m \vdash s : B$$

if and only if

$$x_1 : A_1, \dots, x_n : A_n \vdash A \text{ type} \equiv y_1 : B_1, \dots, y_m : B_m \vdash B \text{ type}$$

and

$$x_1 : A_1, \dots, x_n : A_n \vdash t = s[y_1 \mid x_1, \dots, y_m \mid x_m] : A$$

is derivable.

We define $C(U)$ as the homomorphic image of $R(U)$ induced by this equivalence relation. This means that $C(U)$ is the category of equivalence classes of contexts and of equivalence classes of realizations. Two contexts $x_1 : A_1, \dots, x_n : A_n$ and $y_1 : B_1, \dots, y_m : B_m$ are equivalent if and only if

$$x_1 : A_1, \dots, x_{n-1} : A_{n-1} \vdash A_n \text{ type} \equiv y_1 : B_1, \dots, y_{m-1} : B_{m-1} \vdash B_m \text{ type}$$

and two realizations

$$\begin{aligned} \langle t_1, \dots, t_m \rangle : \langle x_1 : A_1, \dots, x_n : A_n \rangle &\rightarrow \langle y_1 : B_1, \dots, y_m : B_m \rangle \\ \langle s_1, \dots, s_m \rangle : \langle x'_1 : A'_1, \dots, x'_n : A'_n \rangle &\rightarrow \langle y'_1 : B'_1, \dots, y'_m : B'_m \rangle \end{aligned}$$

are equivalent if and only if, for each j , $1 \leq j \leq n$,

$$\begin{aligned} x_1 : A_1, \dots, x_n : A_n \vdash t_j : B_j[y_1 \mid t_1, \dots, y_{j-1} \mid t_{j-1}] &\equiv \\ x'_1 : A'_1, \dots, x'_n : A'_n \vdash s_j : B'_j[y'_1 \mid s_1, \dots, y'_{j-1} \mid s_{j-1}]. \end{aligned}$$

It follows that $C(U)$ is itself a contextual category.

Definition 1.10. If $\mathcal{C}$ and $\mathcal{C}'$ are contextual categories, then a **contextual functor** is a functor $F : \mathcal{C} \rightarrow \mathcal{C}'$ such that:

1. $F(1) = 1$ and if $A \triangleleft B$ in $\mathcal{C}$, then $F(A) \triangleleft F(B)$ in $\mathcal{C}'$;
2. for every object A of $\mathcal{C}$, $F(p(A)) = p(F(A))$;
3. for all f and B such that f^*B is defined in $\mathcal{C}$,

$$F(f^*B) = F(f)^*F(B) \quad \text{and} \quad F(q(f, B)) = q(F(f), F(B)).$$

Con is the category of contextual categories and contextual functors.

1.5 Properties of Contextual Categories

In this section we state some properties about contextual categories that will be used later.

The first useful result is about projections; Although we defined projections on pairs $A \triangleleft B$, we might extend the notion to pairs $A \leq B$ defined as

$$p(B, A) = p(X_1) \circ \dots \circ p(X_n) \circ p(B),$$

where $X_1, \dots, X_n$ is the unique sequence of objects in $\mathcal{C}$ such that $A \triangleleft X_1 \triangleleft \dots \triangleleft X_n \triangleleft B$ in $\mathcal{C}$. In the case where $A = B$, we take $p(B, A) = id_A$. Notation: $B \twoheadrightarrow A$.

In type theory, proving a theorem often requires working under a set of assumptions (a context). In categorical semantics, "assuming a context A " corresponds to taking the relativization of $\mathcal{C}$ to A , noted as $\mathcal{C}_A$. More formally, let $\mathcal{C}$ be a contextual category and A an arbitrary object of $\mathcal{C}$. Then we define the *relativization* of $\mathcal{C}$ to A as the contextual category where:

1. The objects of $\mathcal{C}_A$ are the objects B of $\mathcal{C}$ such that $A \leq B$.

2. For objects B, C in $\mathcal{C}_A$, a morphism from B to C over A is a morphism, i.e., a morphism from B to C in $\mathcal{C}_A$ is a morphism $f: B \rightarrow C$ in $\mathcal{C}$ such that $p(C, A) \circ f = p(B, A)$. Composition of morphisms is inherited from $\mathcal{C}$.
3. $\text{father}_A(B) = \text{father}(B)$ if $A < B$, and $\text{father}_A(A) = A$.
4. For objects $B > A$, $p_A(B) = p(B): B \rightarrow \text{father}(B)$.
5. For B, C, D objects of $\mathcal{C}_A$ with $\text{father}_A(D) = C$ and morphisms $f: B \rightarrow C$, we write

$$f^{*A}D = f^*D \quad \text{and} \quad q_A(f, D) = q(f, D).$$

Now we see that any morphism $f: B \rightarrow A$ in $\mathcal{C}$ induces a pullback functor $f^*: \mathcal{C}_A \rightarrow \mathcal{C}_B$ which is a contextual functor. This functor can be thought of as a way of generalizing knowledge we have in one part of the category to the rest of the category.

Lemma 1.11. *Let $\mathcal{C}$ be a contextual category, and $f: B \rightarrow A$ a morphism in $\mathcal{C}$. Then, for any object C with $C \geq A$, we define an object $f^*C \geq B$ and a morphism $q(f, C): f^*C \rightarrow C$ by induction over $\text{level}C - \text{level}A$:*

1. if $C = A$,

$$f^*C = B \quad \text{and} \quad q(f, C) = f;$$

2. if $C > A$,

$$f^*C = q(f, \text{father}(C))^*C \quad \text{and} \quad q(f, C) = q(q(f, \text{father}(C)), C).$$

Then, if $C \geq A$, the pair $(p(f^*C, B), q(f, C))$ is a pullback cone over $(f, p(C, A))$.

Theorem 1.12. *Let $\mathcal{C}$ be a contextual category and $f: B \rightarrow A$ a morphism in $\mathcal{C}$. By mapping an object C of $\mathcal{C}_A$ to f^*C in $\mathcal{C}_B$, and mapping $g: C \rightarrow D$ over A to $f^*g: f^*C \rightarrow f^*D$ over B determined by*

$$\begin{aligned} p(f^*D, B) \circ f^*g &= p(f^*C, B) \\ q(f, D) \circ f^*g &= q(f, C) \end{aligned}$$

we obtain a contextual functor $f^: \mathcal{C}_A \rightarrow \mathcal{C}_B$ called reindexing along f .*

Theorem 1.13. Let $\mathcal{C}$ be a contextual category and $f: B \rightarrow A$, $g: C \rightarrow B$ be morphisms in $\mathcal{C}$. Then,

$$g^* \circ f^* = (f \circ g)^* \quad \text{and} \quad q(f \circ g, D) = q(f, D) \circ q(g, f^*D)$$

for objects $D \geq A$.

Definition 1.14. Let $\mathcal{C}$ be a contextual category, A and B be objects of $\mathcal{C}$ such that $A \triangleleft B$. The set of **sections** of B is defined as

$$\text{Sections}(B) := \{s: A \rightarrow B \mid p(B) \circ s = id_A\}.$$

If $\text{level}(B) = 1$, then

$$\text{Sections}(B) = \{s \mid s: 1 \rightarrow B\},$$

as any morphism $s: 1 \rightarrow B$ satisfies $p(B) \circ s = id_1$ since 1 is a terminal object. In this case, the notion of section matches the notion of *global element* of B . The concept of section of a context $\langle x_1: A_1, \dots, x_n: A_n, y: B \rangle$ is the categorical equivalent of a term of type B under context $\langle x_1: A_1, \dots, x_n: A_n \rangle$.

1.6 Relevance of the Context Category

The context category is the first example of what it is called a **term model**. This means that it is a category built directly from the syntax of a theory without picking any particular mathematical interpretation. This gives us the *most generic* model of that theory. The three main properties of this models are the following:

1. **Soundness:** Any model of such theory corresponds to a structure preserving functor out of the term model.
2. **Completeness:** A judgement is provable/derivable in the theory if it holds in all models, or equivalently if it holds in the term model.
3. **Initiality:** The term model is often the initial object of the category of all models.

Because of completeness, navigating $C(U)$ is equivalent to manipulating the syntax of U .

We start by proving the completeness of $C(U)$. The following completeness theorem is direct from the definition of $C(U)$:

Theorem 1.15 (Completeness Theorem). Let $\mathcal{M}$ be the canonical interpretation of a theory U in $C(U)$.

1. The T -rule

$$\frac{x_1 : A_1, \dots, x_n : A_n}{A \text{ type}}$$

is derivable if and only if $x_1 : A_1, \dots, x_n : A_n, x : A$ is a context, i.e., if $[x_1 : A_1, \dots, x_n : A_n, x : A]$ in $C(U)$.

2. The $\in$ -rule

$$\frac{x_1 : A_1, \dots, x_n : A_n}{t : A}$$

is derivable if and only if $\langle x_1, \dots, x_n, t \rangle$ is a section of $[x_1 : A_1, \dots, x_n : A_n, x : A]$ in $C(U)$.

3. If the following T -rules are derivable:

$$\Gamma \vdash A_1 \text{ type} \quad \text{and} \quad \Gamma \vdash A_2 \text{ type},$$

then the rule $\Gamma \vdash A_1 = A_2$ is derivable if and only if

$$[\Gamma, x : A_1] = [\Gamma, x : A_2],$$

where x is a variable that does not appear in Γ .

4. If the following $\in$ -rules are derivable:

$$\Gamma \vdash t : A \quad \text{and} \quad \Gamma \vdash s : A,$$

then the rule $\Gamma \vdash t = s : A$ is derivable if, and only if, $\langle x_1, \dots, x_n, t \rangle$ and $\langle x_1, \dots, x_n, s \rangle$ are equal sections of $[\Gamma, x : A]$ in $C(U)$, where x is a variable not appearing in Γ .

Now we prove the initiality of $C(U)$, a special case of Voevodsky's initiality conjecture [12]. Hence, we need to define interpretation of a theory U in a contextual category, and see that the interpretation into $C(U)$ is the initial object of the category of interpretations of U .

Given a contextual category $\mathcal{C}$ and a theory U , an *interpretation* of U in $\mathcal{C}$ is a mapping $\mathcal{M}$ from the T - and $\in$ -rules of U to the $\mathcal{C}$ such that

1. if R is a derived T -rule $\frac{x_1 : A_1, \dots, x_n : A_n}{A \text{ is a type}}$ and we define rules

$R_i \equiv \frac{x_1 : A_1, \dots, x_{i-1} : A_{i-1}}{A_i \text{ is a type}}$ for $1 \leq i \leq n$, then $\mathcal{M}[R_1], \dots, \mathcal{M}[R_n], \mathcal{M}[R]$ are objects of $\mathcal{C}$ such that

$$1 \triangleleft \mathcal{M}[R_1] \triangleleft \dots \triangleleft \mathcal{M}[R_n] \triangleleft \mathcal{M}[R]$$

in $\mathcal{C}$.

2. if R_t is a derived $\in$ -rule $\frac{x_1: A_1, \dots, x_n: A_n}{t: A}$ then $\mathcal{M}[R_t]$ is section of $\mathcal{M}[R]$.

3. if $\Gamma \Rightarrow A = A'$ is a derived T -rule, then if

$$R \equiv \Gamma \Rightarrow A \text{ type} \quad \text{and} \quad R' \equiv \Gamma \Rightarrow A' \text{ type},$$

then $\mathcal{M}[R] = \mathcal{M}[R']$.

4. if $\Gamma \Rightarrow t = t': A$ is a derived $\in$ -rule, then if

$$R_t \equiv \Gamma \Rightarrow t: A, \quad R_{t'} \equiv \Gamma \Rightarrow t': A, \quad R \equiv \Gamma \Rightarrow A \text{ type},$$

then both $\mathcal{M}[R_t]$ and $\mathcal{M}[R_{t'}]$ are sections of $\mathcal{R}$. and $\mathcal{M}[R_t] = \mathcal{M}[R_{t'}]$

There is a canonical interpretation $\mathcal{M}$ of U in $\mathcal{C}(U)$, defined as follows:

1. if $R \equiv x_1: A_1, \dots, x_n: A_n \Rightarrow A$ type is a derived T -judgement, then we define

$$\mathcal{M}[R] = [x_1: A_1, \dots, x_n: A_n, x_{n+1}: A];$$

2. if $R_t \equiv x_1: A_1, \dots, x_n: A_n \Rightarrow t: A$ is a derived $\in$ -judgement, then we define

$$\mathcal{M}[R_t] = [\langle x_1, \dots, x_n, t \rangle]$$

a section of $\mathcal{M}[R]$.

The Completeness Theorem 1.15 guarantees that $\mathcal{M}$ is well defined.

Now we shall define the concept of homomorphism between interpretations of a theory U . Let $\mathcal{C}, \mathcal{C}'$ be two contextual categories, and let $\mathcal{M}_{\mathcal{C}}$ and $\mathcal{M}_{\mathcal{C}'}$ be two U -interpretations in $\mathcal{C}$ and $\mathcal{C}'$ respectively. Then, an interpretation homomorphism is a contextual functor $F: \mathcal{C} \rightarrow \mathcal{C}'$ such that:

1. if R is a T -rule, then $F\mathcal{M}_{\mathcal{C}}[R] = \mathcal{M}_{\mathcal{C}'}[R]$;
2. if R_t is a $\in$ -rule, then $F\mathcal{M}_{\mathcal{C}}[R_t] = \mathcal{M}_{\mathcal{C}'}[R_t]$.

So we can define the category $\mathbf{Con}_U$ as the category with pairs of contextual categories and U -interpretations and their homomorphisms. We can easily see that $(\mathcal{C}(U), \mathcal{M})$ is an initial object of $\mathbf{Con}_U$. Consider an object $(\mathcal{C}, \mathcal{N})$ of $\mathbf{Con}_U$. Let construct a morphism $F: ((\mathcal{C}(U), \mathcal{M})) \rightarrow (\mathcal{C}, \mathcal{N})$, i.e., we have to construct a contextual functor $F: \mathcal{C}(U) \rightarrow \mathcal{C}$ that *commutes* with the interpretation functions. If $[\Gamma]$ an object of $\mathcal{C}(U)$, then there is a T -rule R^Γ such that $\mathcal{M}[R^\Gamma] = \Gamma$. We define $F\Gamma = \mathcal{N}[R^\Gamma]$. In order to define F on the morphisms of $\mathcal{C}(U)$, it suffices to define it for the sections of $\mathcal{C}(U)$, as any morphism

$$\langle t_1, \dots, t_m \rangle: \langle x_1: A_1, \dots, x_n: A_n \rangle \rightarrow \langle y_1: B_1, \dots, y_m: B_m \rangle$$

can be built by composition and pullback functors of sections and projections, and the latter is uniquely determined by the condition that F is a contextual functor. Hence, given a section $s = \langle x_1, \dots, x_n, t \rangle$ of $R \equiv x_1 : A_1, \dots, x_n : A_n, x : A$, by definition there exists a derived $\in$ -rule R_t such that $\mathcal{M}[R_t] = s$, so we can define $Fs = \mathcal{N}[R_t]$. The uniqueness of F is rather simple to see, as we cannot assign any other values to any object or morphism of $C(P)$ without conflicting with the structural conditions of the functor. Hence, $(C(U), \mathcal{M})$ is an initial object of $\mathbf{Con}_U$.

Soundness comes from the initiality of $C(U)$, as any model of U , i.e., any element of $\mathbf{Con}_U$, is equivalent to the initial morphism F we just defined.

1.7 Definition of the Problem

In this section we introduce a formal definition of our problem, proposed at the beginning of the [thesis](#), and reformulate it to an equivalent problem easier to solve with artificial intelligence.

Our problem might be formalized in the following manner: Given a theory U and a derived rule of U of the form

$$\frac{x_1 : A_1, \dots, x_n : A_n}{A \text{ type}}$$

we want to find an expression t of U such that the rule

$$\frac{x_1 : A_1, \dots, x_n : A_n}{t : A}$$

is a derived rule of U .

This problem is undecidable, in general, as it follows from the undecidability of the inhabitation in a *GAT*. This is why we will not try to find an algorithm that either finds a term or says that the type is empty. We aim to find an algorithm that, in case that the type is inhabited, it finds a term in a finite number of steps. In case that the type is empty, it does not have to give an output.

We can consider this problem from the point of view of the context category $C(U)$, where by the Completeness Theorem 1.15 it is rephrased as follows: Given a context $\langle x_1 : A_1, \dots, x_n : A_n \rangle$ in $C(U)$, we want to find a section of $\langle x_1 : A_1, \dots, x_n : A_n, x : A \rangle$ in $C(U)$.

By naming $\Gamma \equiv \langle x_1 : A_1, \dots, x_n : A_n \rangle$, we can restrict our search to the *relativization* of $C(U)$ to $[\Gamma]$, that we denote as $\mathcal{C}_\Gamma$. We denote a morphism $\langle x_1, \dots, x_n, t_{n+1}, \dots, t_{n+m} \rangle$ in $\mathcal{C}_\Gamma$, as $\langle t_{n+1}, \dots, t_{n+m} \rangle_\Gamma$.

1.8 Expansion Algorithm

In this section, we define the *expansion algorithm* that will be responsible of finding $\langle t \rangle_\Gamma$ in $\mathcal{C}_\Gamma$. The complexity of this algorithm is really high, and it will serve as a baseline to be optimized by a graph neural network. We will explain this in the last chapter.

First, consider a finite subcategory G of $\mathcal{C}_A$ which contains $\Gamma, \Gamma, x: A$ and its projection. The process consists of the construction of a sequence $\{G_i\}_i$ of finite subcategories of $\mathcal{C}_\Gamma$ such that $G_0 = G$, and that G_{n+1} is computed by the process described below from G_n , and where $G_n \subseteq G_{n+1}$ for all $n \geq 0$. We say that an object or morphism of $\mathcal{C}_A$ is *constructible by expansion* if there exists an $n \geq 0$ such that the given object or morphism is contained in G_n . The process of constructing G_{n+1} from G_n consists of applying the following rules to each context $\Gamma y_1: B_1, \dots, y_m: B_m$ of G_n :

E1 For each sort symbol S with a well-formed introductory rule

$$\frac{z_1: C_1, \dots, z_n: C_w}{A(z_1, \dots, z_w) \text{ type}}$$

such that G_n contains sections

$$\begin{aligned} \Gamma, y_1: B_1, \dots, y_m: B_m &\xrightarrow{\langle y_1, \dots, y_m, t_1 \rangle_\Gamma} \Gamma, y_1: B_1, \dots, y_m: B_m, z_1: C_1 \\ \Gamma, y_1: B_1, \dots, y_m: B_m &\xrightarrow{\langle y_1, \dots, y_m, t_2 \rangle_\Gamma} \Gamma, y_1: B_1, \dots, y_m: B_m, z_2: C_2[z_1|t_1] \\ &\vdots \\ \Gamma, y_1: B_1, \dots, y_m: B_m &\xrightarrow{\langle y_1, \dots, y_m, t_w \rangle_\Gamma} \Gamma, y_1: B_1, \dots, y_m: B_m, z_w: C_w[z_1|t_1, \dots, z_{w-1}|t_{w-1}] \end{aligned}$$

then we add the context $\Gamma, y_1: B_1, \dots, y_m: B_m, y: A(t_1, \dots, t_w)$ and its projection to G_{n+1} .

E2 Add the morphisms

$$\Gamma, y_1: B_1, \dots, y_m: B_m \xrightarrow{\langle y_1, \dots, y_m, y_i \rangle_\Gamma} \Gamma, y_1: B_1, \dots, y_m: B_m, y: B_i$$

and their corresponding projections for each $0 \leq i \leq n$.

E3 For every operator symbol F with a well-formed introductory rule

$$\frac{z_1: C_1, \dots, z_w: C_w}{F(z_1, \dots, z_w): C}$$

if the following sections exist in G_n

$$\begin{aligned} \Gamma, y_1 : B_1, \dots, y_m : B_m &\xrightarrow{\langle y_1, \dots, y_m, t_1 \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_1 : C_1 \\ \Gamma, y_1 : B_1, \dots, y_m : B_m &\xrightarrow{\langle y_1, \dots, y_m, t_2 \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_2 : C_2[z_1 | t_1] \\ &\vdots \\ \Gamma, y_1 : B_1, \dots, y_m : B_m &\xrightarrow{\langle y_1, \dots, y_m, t_w \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_w : C_w[z_1 | t_1, \dots, z_{w-1} | t_{w-1}] \end{aligned}$$

then we add the section

$$\Gamma, y_1 : B_1, \dots, y_m : B_m \xrightarrow{\langle y_1, \dots, y_m, F(t_1, \dots, t_w) \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z : C[z_1 | t_1, \dots, z_w | t_w]$$

E4 For every morphism

$$\langle t_1, \dots, t_{m-1} \rangle_\Gamma : \Gamma, z_1 : C_1, \dots, z_w : C_w \rightarrow \Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}$$

we construct the corresponding pullback

$$\begin{array}{ccc} \Gamma, z_1 : C_1, \dots, z_w : C_w, y_m : B_m[t_1 | y_1, \dots, t_{m-1} | y_{m-1}] & \xrightarrow{\langle t_1, \dots, t_{m-1}, y_m \rangle_\Gamma} & \Gamma, y_1 : B_1, \dots, y_m : B_m \\ \downarrow & & \downarrow \\ \Gamma, z_1 : C_1, \dots, z_w : C_w & \xrightarrow{\langle t_1, \dots, t_{m-1} \rangle_\Gamma} & \Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1} \end{array}$$

in G_{n+1} .

Theorem 1.16. *Every object and every morphism of $\mathcal{C}_A$ are constructible by expansion.*

Proof. We prove this result by structural induction over the underlying derived rules corresponding to such object (context) or morphism (realization).

Consider a context $\Gamma, y_1 : B_1, \dots, y_m : B_m$ of U , which is a representative of an object in $C(U)$. This means, by the definition of context, that its underlying rule

$$\frac{\Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}}{B_m \text{ type}} \quad (1.1)$$

is a derived rule of U and y_m is a variable different to $x_1, \dots, x_n, y_1, \dots, y_{m-1}$. Hence, the premise $\Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}$ i.e., there exists $n_0 \geq 0$ such that $\Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}$ is an object G_{n_0} . As (1.1) is a derived rule, it must have been derived by the only derivation rule that has a T -rule as a conclusion:

CF2(a). Hence, there exists a sort symbol S with a well formed introductory rule $\frac{z_1 : C_1, \dots, z_w : C_w}{S(z_1, \dots, z_w)}$ type and derived rules

$$\begin{array}{c} \frac{\Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}}{t_1 : C_1}, \\ \frac{\Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}}{t_2 : C_2[t_1|z_1]}, \\ \vdots \\ \frac{\Gamma y_1 : B_1, \dots, y_{m-1} : B_{m-1}}{t_w : C_w[t_1|z_1, \dots, t_{w-1}|z_{w-1}]} \end{array}$$

By induction hypothesis, their corresponding sections

$$\begin{array}{c} \langle y_1, \dots, y_{m-1}, t_j \rangle_\Gamma : \Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1} \rightarrow \\ \Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}, z_j : C_j[t_1|z_1, \dots, t_{j-1}|z_{j-1}] \end{array}$$

are constructible by expansion, so for each j there exists some n_j such that the section exists in G_{n_j} .

Take $n = \max\{n_0, n_1, \dots, n_w\}$. Then, all of the objects and sections mentioned belong to G_n , so, by the expansion rule E1,

$$\Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1} \leftarrow \Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}, y_m : S(t_1, \dots, t_w)$$

is included in G_{n+1} . And by the equivalence of intended equivalence of denotation

$$\Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}, y_m : S(t_1, \dots, t_w) \equiv \Gamma, y_1 : B_1, \dots, y_m : B_m$$

so $\Gamma, y_1 : B_1, \dots, y_m : B_m$ is constructible by expansion. Now consider a realization

$$\Gamma, y_1 : B_1, \dots, y_m : B_m \xrightarrow{\langle t_1, \dots, t_w \rangle_\Gamma} \Gamma z_1 : C_1, \dots, z_w : C_w$$

representative of a morphism in $\mathcal{C}_A$. By induction hypothesis, we know that both $\Gamma, y_1 : B_1, \dots, y_m : B_m$ and $\Gamma, z_1 : C_1, \dots, z_w : C_w$ are constructible by expansion.

Moreover, as $\langle t_1, \dots, t_w \rangle_\Gamma$ is an operator, for each $1 \leq j \leq w$, the rule

$$\frac{\Gamma, y_1 : B_1, \dots, y_m : B_m}{t_j : C_j[z_1 \mid t_1, \dots, z_{j-1} \mid t_{j-1}]} \quad (1.2)$$

is a derived rule of U . Now we see that the following sections are constructible by expansion:

$$\begin{array}{c} \Gamma, y_1 : B_1, \dots, y_m : B_m \xrightarrow{\langle y_1, \dots, y_m, t_1 \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_1 : C_1 \\ \Gamma, y_1 : B_1, \dots, y_m : B_m \xrightarrow{\langle y_1, \dots, y_m, t_2 \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_2 : C_2[t_1|z_1] \\ \vdots \\ \Gamma, y_1 : B_1, \dots, y_m : B_m \xrightarrow{\langle y_1, \dots, y_m, t_w \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_w : C_w[t_1|z_1, \dots, t_{w-1}|z_{w-1}]. \end{array}$$

To do this, we use the way any rule (1.2) is derived. This can be done by one of the following rules: T1, CF1, CF2(b). Hence,

- if derived by T1, we have that both

$$\frac{\Gamma, y_1 : B_1, \dots, y_m : B_m}{A = C_j[z_1 \mid t_1, \dots, z_{j-1}]} \quad \text{and} \quad \frac{\Gamma, y_1 : B_1, \dots, y_m : B_m}{t_j : A}$$

are derived rules of U . So by the first rule, due to the equivalence relation of intended equality of denotation, we know that

$$\Gamma, y_1 : B_1, \dots, y_m : B_m, z_j : A \equiv \Gamma, y_1 : B_1, \dots, y_m : B_m, z_j : C_j[t_1|z_1, \dots, t_{j-1}|z_{j-1}]$$

hence they are the same object in C_Γ . And by the second one, applying induction, we have that

$$\Gamma, y_1 : B_1, \dots, y_m : B_m \xrightarrow{\langle y_1, \dots, y_m, t_j \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_j : A$$

is constructible by expansion, so as $\Gamma, y_1 : B_1, \dots, y_m : B_m, z_j : A$ and $\Gamma, y_1 : B_1, \dots, y_m : B_m, z_j : C_j[t_1|z_1, \dots, t_{j-1}|z_{j-1}]$ are the same object in C_A , then

$$\Gamma, y_1 : B_1, \dots, y_m : B_m \xrightarrow{\langle y_1, \dots, y_m, t_j \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_j : C_j[t_1|z_1, \dots, t_{j-1}|z_{j-1}]$$

is constructible by expansion.

- If derived by CF1, we have that

$$\frac{\Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}}{B_m \text{ type}}$$

is a derived rule of U , where y_m is a variable distinct from all $x_1, \dots, x_n, y_1, \dots, y_{m-1}$, and that there exists an $i, 1 \leq i \leq m$, such that

$$\frac{\Gamma, y_1 : B_1, \dots, y_m : B_m}{t_j : C[z_1 \mid t_1, \dots, z_{j-1} \mid t_{j-1}]} \equiv \frac{\Gamma, y_1 : B_1, \dots, y_m : B_m}{y_i : B_i}.$$

Then, by induction hypothesis, the context $\Gamma, y_1 : B_1, \dots, y_m : B_m$ is constructible by expansion. By applying the expansion rule E2 and an argument equivalent to the one above, we obtain that

$$\Gamma, y_1 : B_1, \dots, y_m : B_m \xrightarrow{\langle y_1, \dots, y_m, t_j \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_j : C_j[z_1 \mid t_1, \dots, z_{j-1} \mid t_{j-1}]$$

is constructible by expansion.

- If derived by CF2(b), there exists an operator symbol F with a well-formed introductory rule

$$\frac{v_1 : D_1, \dots, v_k : D_k}{F(v_1, \dots, v_k) : D}$$

and derived rules

$$\begin{array}{c} \frac{\Gamma, y_1 : B_1, \dots, y_m : B_m}{s_1 : D_1} \\ \frac{\Gamma, y_1 : B_1, \dots, y_m : B_m}{s_2 : D_2[v_1 | s_1]} \\ \vdots \\ \frac{\Gamma, y_1 : B_1, \dots, y_m : B_m}{s_v : D_k[v_1 | s_1, \dots, v_{k-1} | s_{k-1}]} \end{array}$$

such that

$$\begin{array}{c} \frac{\Gamma, y_1 : B_1, \dots, y_m : B_m}{D[s_1 | c_1, \dots, s_v | c_v] = C_j[t_1 | z_1, \dots, t_{j-1} | z_{j-1}]} \\ \frac{\Gamma, y_1 : B_1, \dots, y_m : B_m}{t_j = F(s_1, \dots, s_v) : C_j[t_1 | z_1, \dots, t_{j-1} | z_{j-1}]} \end{array}$$

are derived rules of U . By the equivalence relation of identity of denotation, we have to prove that the morphism

$$\Gamma, y_1 : B_1, \dots, y_m : B_m \xrightarrow{\langle y_1, \dots, y_m, F(s_1, \dots, s_k) \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_j : D[s_1 | c_1, \dots, s_k | v_k]$$

is constructible by expansion. By induction, we have that the following morphisms are constructible by expansion:

$$\begin{array}{c} \Gamma, y_1 : B_1, \dots, y_m : B_m \xrightarrow{\langle y_1, \dots, y_m, s_1 \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, v_1 : D_1 \\ \Gamma, y_1 : B_1, \dots, y_m : B_m \xrightarrow{\langle y_1, \dots, y_m, s_2 \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, v_2 : D_2[s_1 | v_1] \\ \vdots \\ \Gamma, y_1 : B_1, \dots, y_m : B_m \xrightarrow{\langle y_1, \dots, y_m, s_v \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, c_v : D_k[s_1 | v_1, \dots, s_{k-1} | v_{k-1}] \end{array}$$

Hence, by applying the expansion rule E3, we have that

$$\Gamma, y_1 : B_1, \dots, y_m : B_m \xrightarrow{\langle y_1, \dots, y_m, F(s_1, \dots, s_k) \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_j : D_j[s_1 | v_1, \dots, s_k | c_k]$$

is constructible by expansion, as we wanted.

Now we show that the morphism

$$\Gamma, y_1 : B_1, \dots, y_m : B_m \rangle \xrightarrow{\langle y_1, \dots, y_m, t_1, \dots, t_w \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_1 : C_1, \dots, z_w : C_w$$

is constructible by expansion. To do this, we prove that, given j with $2 < j \leq n$, if

$$\Gamma, y_1 : B_1, \dots, y_m : B_m \rangle \xrightarrow{\langle y_1, \dots, y_m, t_1, \dots, t_{j-1} \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_1 : C_1, \dots, z_{j-1} : C_{j-1}$$

$$\Gamma, y_1 : B_1, \dots, y_m : B_m \rangle \xrightarrow{\langle y_1, \dots, y_m, t_j \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_j : C_j[z_1 \mid \dots z_{j-1} \mid t_{j-1}]$$

are constructible by expansion so is

$$\Gamma, y_1 : B_1, \dots, y_m : B_m \rangle \xrightarrow{\langle y_1, \dots, y_m, t_1, \dots, t_j \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_1 : C_1, \dots, z_j : C_j$$

By expansion rule E4, the following diagram is constructible by expansion:

$$\begin{array}{ccc} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_j : C_j[t_1|z_1, \dots, t_j|z_j] \xrightarrow{\langle y_1, \dots, y_m, t_1, \dots, t_{j-1}, z_j \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_1 : C_1, \dots, z_j : C_j & & \\ \downarrow & & \downarrow \\ \Gamma, y_1 : B_1, \dots, y_m : B_m \rangle \xrightarrow{\langle y_1, \dots, y_m, t_1, \dots, t_{j-1} \rangle_\Gamma} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_1 : C_1, \dots, z_{j-1} : C_{j-1} & & \end{array}$$

and by composition we get that

$$\langle y_1, \dots, y_m, t_1, \dots, t_{j-1}, z_j \rangle_\Gamma \circ \langle y_1, \dots, y_m, t_j \rangle_\Gamma = \langle y_1, \dots, y_m, t_1, \dots, t_j \rangle_\Gamma$$

is constructible by expansion. It remains to see that the morphism

$$\Gamma, y_1 : B_1, \dots, y_m : B_m, z_1 : C_1, \dots, z_w : C_w \rangle \xrightarrow{\langle z_1, \dots, z_w \rangle_\Gamma} \Gamma, z_1 : C_1, \dots, z_w : C_w$$

is constructible by expansion. That comes directly from the fact that $\Gamma, z_1 : C_1, \dots, z_w : C_w$ is a context, and repeated application of the rule E4 gives us the following diagram:

$$\begin{array}{ccc} \Gamma, y_1 : B_1, \dots, y_m : B_m, z_1 : C_1, \dots, z_w : C_w \rangle & \xrightarrow{\langle z_1, \dots, z_w \rangle_\Gamma} & \Gamma, z_1 : C_1, \dots, z_w : C_w \\ \downarrow & & \downarrow \\ \Gamma, y_1 : B_1, \dots, y_m : B_m, z_1 : C_1, \dots, z_{w-1} : C_{w-1} \rangle & \xrightarrow{\langle z_1, \dots, z_{w-1} \rangle_\Gamma} & \Gamma, z_1 : C_1, \dots, z_{w-1} : C_{w-1} \\ \downarrow & & \downarrow \\ \vdots & & \vdots \\ \downarrow & & \downarrow \\ \Gamma, y_1 : B_1, \dots, y_m : B_m, z_1 : C_1 \rangle & \xrightarrow{\langle z_1 \rangle_\Gamma} & \Gamma, z_1 : C_1 \\ \downarrow & & \downarrow \\ \Gamma, y_1 : B_1, \dots, y_m : B_m \rangle & \xrightarrow{\langle \rangle_\Gamma} & \Gamma \end{array}$$

This completes the proof of Theorem 1.16. $\square$

Chapter 2

The Calculus of Constructions

In Chapter 1, we established a pipeline for autonomous theorem proving: take a theory, translate its syntax into a Contextual Category, and reframe proof-finding as the geometric problem of finding a section in a graph.

In this chapter, we scale up the complexity to the Calculus of Constructions (CoC). Introduced by Coquand and Huet [5], CoC is a powerful high-order typed lambda calculus. It serves as the foundation for the Calculus of Inductive Constructions, the type theory powering modern proof assistants like Rocq (formerly Coq).

The primary additions in CoC, compared to Generalized Algebraic Theories (GATs), are:

1. **Product Types (Π -types):** These are first-class citizens, meaning dependent function types exist at the base level rather than just in the theory's signature.
2. **The Universe of Propositions (Prop):** A type universe whose terms are themselves types, allowing for higher-order logic and universal quantification over types.

We define syntax of CoC, map it to a richer categorical structure called Doctrine of Constructions, and extend our Expansion Algorithm to navigate this new, highly complex search space.

2.1 Syntactic Formulation

In this section, we formally define the syntax of Calculus of Constructions. We use the approach of Streicher in [11], since the original definition in [5] is characterized by its syntax overloading, making it much harder to follow. They just use $[x : A]B$ to present product types, functional abstraction and universal

quantification. Also, properties are presented as terms of type Prop and as types themselves. To solve the latter, we will be using a type constructor Proof that, given a proposition $p : \text{Prop}$, yields $\text{Proof}(p)$ the type of proofs of proposition p .

First, we define the class of *pre-well-formed type expressions*, ranged over by capital letters, and the class of *pre-well-formed object expressions*, ranged over by lower case letters, by mutual recursion in a BMF-like way:

$$\begin{aligned}
 E ::= & \quad (\Pi x : E) E && \text{product of types} \\
 & \quad | \text{Prop} && \text{type of propositions} \\
 & \quad | \text{Proof}(t) && \text{type of proofs of proposition } t \\
 \\
 t ::= & \quad x && \text{variable} \\
 & \quad | (\lambda x : E) t && \text{functional abstraction} \\
 & \quad | \text{App}([x : E]E, t, t) && \text{function application} \\
 & \quad | (\forall x : E) t && \text{universal quantification}
 \end{aligned}$$

We say a *pre-context* is a syntactical expression of the form $x_1 : A_1, \dots, x_n : A_n$, where $x_1, \dots, x_n$ are pairwise syntactically different variables. Contexts are ranged over by capital greek letters. We define $\text{Var}(\Gamma)$ as the set of variables declared in Γ . We will not distinguish expressions which are equivalent up to renaming of variables (α -conversion), to do this we will be using *deBruijn indices*¹. We define a *pre-judgment* to be a syntactic expression of one of the following forms:

$$\begin{aligned}
 A \text{ type} & \quad A \text{ is a type} \\
 A = B & \quad A \text{ and } B \text{ are equal types} \\
 t : A & \quad t \text{ is an object of type } A \\
 t = s : A & \quad t \text{ and } s \text{ are equal objects of type } A
 \end{aligned}$$

we will use the letter $\mathcal{J}$ to refer to pre-judgments. We define a *pre-sequent* as a syntactic expression of the following form:

$$\begin{aligned}
 \Gamma \text{ ok} & \quad \Gamma \text{ is a context} \\
 \Gamma = \Delta & \quad \Gamma \text{ and } \Delta \text{ are equal contexts} \\
 \Gamma \vdash \mathcal{J} & \quad \text{the judgement } \mathcal{J} \text{ holds w.r.t. } \Gamma
 \end{aligned}$$

Finally, we define the set of *sequents*, i.e., valid pre-sequents, to be generated by the set of rules in A.2. Sequents are ranged over by capital italic letters. This set of rules is more extent that the original in order to make explicit rules for handling contexts and to add more rules stating equality of types and objects. We can distinguish different kinds of rules for different purposes:

¹*deBruijn indices* are a method to refer to declarations not by name (variables) but by numbers which tell how far one has to go back into the current context of an expression to find the declaration of the thing it refers to.

1. Context Rules: their purpose is to make contexts first-class citizens of CoC, unlike in GATs where they were defined outside the rule system. These includes: context formation rules and context equality rules
2. Structural Rules for Judgments: their purpose is to form and serve as equality rules for judgments.
3. Rules for Product Types: these rules include everything needed for working with products.
4. Rules for Propositions: finally, the machinery to work with the Prop universe, Proof and $\forall$.

2.2 Doctrines of Constructions

We leverage the concept of context categories introduced in the earlier chapter to serve as a basic structure for handling dependent types, and add more conditions so that it can handle products of families of types, universal quantification and the type of properties, in order to develop a categorical structure that captures the interpretation of the Calculus of Constructions.

Although products might be added directly in the GAT structure, we enforce them to the base level as they are needed to the definition of universal quantification.

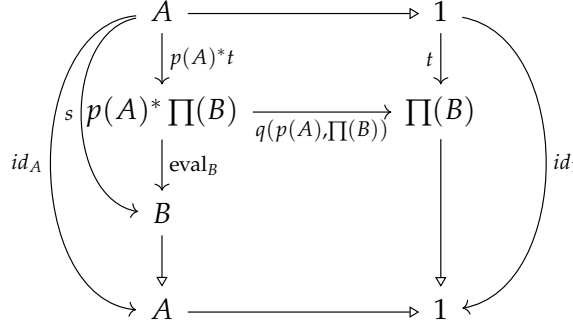
Definition 2.1. A contextual category with products of families of types is a category $\mathcal{C}$ together with two mappings Π and eval defined over the collection of objects B of $\mathcal{C}$ with $\text{level}(B) \geq 2$, such that:

1. Whenever $1 \triangleleft A \triangleleft B$, then $1 \triangleleft \Pi(B)$ and $\text{eval}_B: p(A)^* \Pi(B) \rightarrow B$ is a morphism over A

$$\begin{array}{ccc}
 p(A)^* \Pi(B) & \xrightarrow{\text{eval}_B} & B \\
 \searrow p(p(A)^* \Pi(B)) & & \swarrow p(B) \\
 & A &
 \end{array}$$

such that for all $s \in \text{Sections}(B)$ there is a unique section $t \in \text{Sections}(\Pi(B))$

with $\text{eval}_B \circ p(A)^*t = s$. This unique section t is called $\text{curry}(s)$.



Observation 2.2. as any section t of $\Pi(B)$ gives a section $\text{eval}_B \circ p(A)^*t$ of B , hence we have a canonical 1-1-correspondence of $\text{Sections}(B)$ and $\text{Sections}(\Pi(B))$.

2. For any object J and any morphism $f : J \rightarrow 1$, the following conditions are satisfied:

$$f^*(\Pi(B)) = \Pi(f^*B) \quad \text{and} \quad f^*(\text{eval}_B) = \text{eval}_{f^*B}.$$

It follows from the definition that substitution preserves functional abstraction or currying:

Lemma 2.3. Let $\mathcal{C}$ be a contextual category with products of families of types. Let $1 \triangleleft A \triangleleft B$ be objects and $f : J \rightarrow 1$ a morphism in $\mathcal{C}$. If t is a section of B , then

$$f^*(\text{curry}(B)) = \text{curry}(f^*B).$$

Notice that in any contextual category with products of families of types, the following equations hold:

$$\text{eval}_B \circ p(A)^*\text{curry}(s) = s \tag{\beta}$$

$$\text{curry}(\text{eval}_B \circ p(A)^*t) = t \tag{\eta}$$

The first equation corresponds to the β -rule of typed λ -calculi and the second equation corresponds to the η -rule of typed λ -calculi. Contextual Categories with products correspond to λ -calculus with dependent types.

Definition 2.4. Let $\mathcal{C}, \mathcal{C}'$ be two contextual categories with products of families of types, we a contextual functor $F : \mathcal{C} \rightarrow \mathcal{C}'$ preserves products of families of types if for $1 \triangleleft A \triangleleft B$ in $\mathcal{C}$,

$$F \Pi(B) = \Pi(FB) \quad \text{and} \quad F \text{eval}_B = \text{eval}_{FB}$$

NOTE: this definition is better for actually proving relativization on the section on Problem Redefinition. An alternative definition of contextual categories with products of families of types is given by Voevodsky in this [INSERT REF](#)

Definition 2.5. Let $\mathcal{C}$ be a category. Let $g: Z \rightarrow Y$, $f: Y \rightarrow X$ be a pair of morphisms such that for any $U \rightarrow X$ there exists a pullback $U \times_X Y$. A pair

$$(w: W \rightarrow X, h: W \times_X Y \rightarrow Z)$$

such that $g \circ h = pr$ is called a universal pair for (f, g) if for any $U \rightarrow X$ the map

$$\begin{aligned} \text{Hom}_X(U, W) &\longrightarrow \text{Hom}_Y(U \times_X Y, Z) \\ u &\longmapsto h \circ (u \times id_Y) \end{aligned}$$

is a bijection.

Definition 2.6. A contextual category with products of families of types (or a Π -contextual category in the reference) consist of a contextual category $\mathcal{C}$ together with:

1. for each object Y of $\mathcal{C}$, with $\text{level}(Y) \geq 2$, an object $\Pi(Y)$ of $\mathcal{C}$ such that $\text{father}(\Pi(Y)) = \text{father}^2(Y)$;
2. for each object Y of $\mathcal{C}$, with $\text{level}(Y) \geq 2$, a morphism

$$\text{eval}_Y: p(\text{father}(Y))^* \Pi(Y) \rightarrow Y$$

over $\text{father}(Y)$ such that:

- (a) for any $f: Z \rightarrow \text{father}^2(Y)$

$$f^* \Pi(Y) = \Pi(f^* Y) \quad \text{and} \quad f^* \text{eval}_Y = \text{eval}_{f^* Y},$$

- (b) $(\Pi(Y), \text{eval}_Y)$ is a universal pair for $(p(Y), p(\text{father}(Y)))$.

Now we add the type of propositions. To do it, we find objects in our category whose sections, i.e., its terms, correspond to objects.

Definition 2.7. Let $\mathcal{C}$ be a context category. An *enumeration of types* of $\mathcal{C}$ is an object T of $\mathcal{C}$ with $\text{level}(T) = 2$.

An enumeration of types T of $\mathcal{C}$ is called a *collection of types* of $\mathcal{C}$ if for all objects A and morphisms $f, g: A \rightarrow \text{father}(T)$,

$$f^* T = g^* T \implies f = g.$$

It is the object $\text{father}(T)$ that we are interested in.

And finally, by wrapping everything up and adding universal quantification we have the desired structure:

Definition 2.8. A *doctrine of constructions* is a contextual category with products of families of types together with a collection of types T such that for all $A, B \in \text{Ob}\mathcal{C}$ with $A \triangleleft B$ and $f: B \rightarrow \text{Prop}$, where $\text{Prop} = \text{father}T$, there exists a necessarily unique morphism $g: A \rightarrow \text{Prop}$ also denoted by $\forall(f)$ such that

$$g^*T = \forall(f)^*T = \prod(f^*T).$$

Definition 2.9. Let $\mathcal{C}, \mathcal{C}'$ be two doctrines of constructions. A *functor of doctrines of constructions* is a contextual functor $F: \mathcal{C} \rightarrow \mathcal{C}'$ that preserves products of families of types such that if T and T' are the selected collections of types of $\mathcal{C}$ and $\mathcal{C}'$ respectively, $FT = T'$, and for any $A \triangleleft B$ in $\mathcal{C}$, and $f: B \rightarrow \text{Prop}$, $F(\forall(f)) = \forall(F(f))$.

We consider the category **DoC** of doctrines of constructions and functors of doctrines of constructions.

Now we see that doctrines of constructions correspond to models of Calculus of Constructions.

2.3 Interpretation of Calculus of Constructions

In this section we define the interpretation of Calculus of Constructions in arbitrary doctrines of constructions.

To do this, fixed one doctrine of constructions, an **interpretation function** $\mathcal{M}$ should be a mapping such that:

1. for every pre-context Γ of length n , if $\mathcal{M}[\Gamma]$ is defined, then $\mathcal{M}[\Gamma]$ is an object of level n ;
2. for every pre-context of length n and type expression A , if $\mathcal{M}[\Gamma; A]$ is defined then $\mathcal{M}[\Gamma; A]$ is an object of level $n + 1$ and $\mathcal{M}[\Gamma] \triangleleft \mathcal{M}[\Gamma; A]$;
3. for every pre-context Γ of length n and object expression t , if $\mathcal{M}[\Gamma; t]$ is defined then it is a morphism $s: \mathcal{M}[\Gamma] \rightarrow A$ where A is an object of level $n + 1$ such that

$$\mathcal{M}[\Gamma] \triangleleft A \quad \text{and} \quad p(A) \circ s = \text{id}_{\mathcal{M}[\Gamma]}.$$

We want to build this function by induction. We need an auxiliary definition:

Definition 2.10. By structural induction of expressions we define a function **depth** mapping pre-well-formed expressions and pre-contexts to ω :

$$\text{depth}[x] = 1$$

$$\text{depth}[\text{Prop}] = 1$$

$$\text{depth}[\text{Proof}(A)] = \text{depth}[A] + 1$$

$$\text{depth}[(\prod x : A)B] = \text{depth}[A] + \text{depth}[B] + 1$$

$$\text{depth}[(\lambda x : A)t] = \text{depth}[A] + \text{depth}[t] + 1$$

$$\text{depth}[(\forall x : A)t] = \text{depth}[A] + \text{depth}[t] + 1$$

$$\text{depth}[\text{App}([x : A]B, t, s)] = \text{depth}[A] + \text{depth}[B] + \text{depth}[t] + \text{depth}[s] + 1$$

$$\text{For } \Gamma \equiv x_1 : A_1, \dots, x_n : A_n, \quad \text{depth}[\Gamma] = \sum_{i=1}^n \text{depth}[A_i]$$

We define $\mathcal{M}$ by induction on the level of its arguments, where

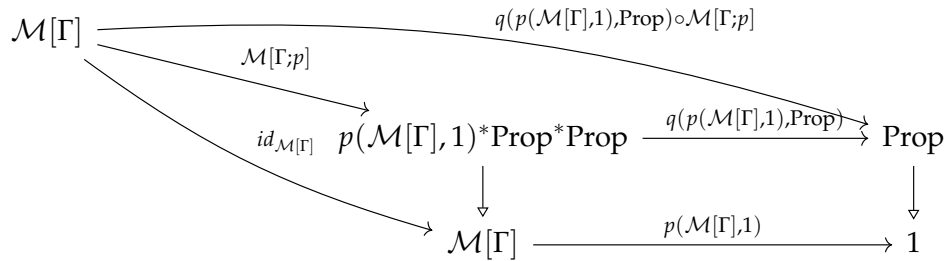
$$\text{level}[\Gamma] = \text{depth}[\Gamma],$$

$$\text{level}[\Gamma; A] = \text{depth}[\Gamma] + \text{depth}[A],$$

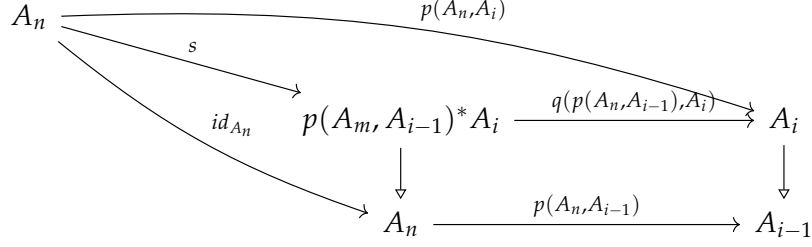
$$\text{level}[\Gamma; t] = \text{depth}[\Gamma] + \text{depth}[t]$$

by the following clauses:

1. $\mathcal{M}[] = 1$;
2. $\mathcal{M}[\Gamma, x : A] = \mathcal{M}[\Gamma; A]$;
3. $\mathcal{M}[\Gamma, (\prod x : A)B] = \prod(\mathcal{M}[\Gamma, x : A; B])$ if $\mathcal{M}[\Gamma, x : A; B]$ is defined;
4. $\mathcal{M}[\Gamma, \text{Prop}] = p(\mathcal{M}, 1)^* \text{Prop}$ if $\mathcal{M}[\Gamma]$ is defined;
5. $\mathcal{M}[\Gamma; \text{Proof}(p)] = \mathcal{M}[\Gamma; p]^* p(\mathcal{M}[\Gamma], 1)^* T$ if $\mathcal{M}[\Gamma; p]$ is defined and a section of $p(\mathcal{M}[\Gamma], 1)^* \text{Prop}$



6. $\mathcal{M}[\Gamma, x] = s$ provided that $\Gamma \equiv x_1 : A_1, \dots, x_n : A_n$ and $x_i \equiv x$ and $\mathcal{M}[\Gamma]$ is defined and $\mathcal{M}[\Gamma] = A_n \triangleright \dots \triangleright A_1 \triangleright A_0 \triangleright 1$ and s as follows



or, in other terms, $\mathcal{M}[\Gamma; x_i] = p(A_n, A_i)^* \Delta(A_i)$ where $\Delta(A_i)$ is the unique arrow $m : A_i \rightarrow p(A_i)^* A_i$ such that

$$p(p(A_i)^* A_i) \circ m = q(p(A_i), A_i) \circ m = id_{A_i};$$

7. $\mathcal{M}[\Gamma; (\lambda x : A) t] = \text{curry}(\mathcal{M}[\Gamma, x : A; t])$ if $\mathcal{M}[\Gamma, x : A; t]$ is defined. By induction hypothesis we have that $\mathcal{M}[\Gamma] \triangleleft \mathcal{M}[\Gamma; A] \triangleleft \text{cod}(\mathcal{M}[\Gamma, x : A; t])$ and s is a section of $\mathcal{M}[\Gamma, x : A; t]$ and therefore we have:

$$\text{eval}_{\text{cod}(\mathcal{M}[\Gamma, x : A; t])} \circ p(\mathcal{M}[\Gamma; A])^* \mathcal{M}[\Gamma; (\lambda x : A) t] = \mathcal{M}[\Gamma, x : A; t];$$

8. $\mathcal{M}[\Gamma; \text{App}([x : A] B), t, s] = \mathcal{M}[\Gamma; s]^* (\text{eval}_{\mathcal{M}[\Gamma, x : A; B]} \circ p(\mathcal{M}[\Gamma; A])^* \mathcal{M}[\Gamma; t])$ provided that $\mathcal{M}[\Gamma; t]$, $\mathcal{M}[\Gamma; s]$, $\mathcal{M}[\Gamma, x : A; B]$ are defined, and $\mathcal{M}[\Gamma; s]$ is a section of $\mathcal{M}[\Gamma; A]$ and $\mathcal{M}[\Gamma; t]$ is a section of $\prod(\mathcal{M}[\Gamma, x : A; B])$

$$\begin{array}{ccc}
 \mathcal{M}[\Gamma] & \xrightarrow{\mathcal{M}[\Gamma; s]} & \mathcal{M}[\Gamma; A] \\
 \mathcal{M}[\Gamma; s]^* (\text{eval}_{\mathcal{M}[\Gamma, x : A; B]} \circ p(\mathcal{M}[\Gamma; A])^* \mathcal{M}[\Gamma; t]) \downarrow & & \downarrow \text{eval}_{\mathcal{M}[\Gamma, x : A; B]} \circ p(\mathcal{M}[\Gamma; A])^* \mathcal{M}[\Gamma; t] \\
 \mathcal{M}[\Gamma; s]^* \mathcal{M}[\Gamma, x : A; B] & \xrightarrow{q(\mathcal{M}[\Gamma; s], \mathcal{M}[\Gamma, x : A; B])} & \mathcal{M}[\Gamma, x : A; B] \\
 \downarrow & & \downarrow \\
 \mathcal{M}[\Gamma] & \xrightarrow{\mathcal{M}[\Gamma; s]} & \mathcal{M}[\Gamma; A]
 \end{array}$$

9. $\mathcal{M}[\Gamma; (\forall x : A) p] = s$ provided $\mathcal{M}[\Gamma, x : A; p]$ is a section of $p(\mathcal{M}[\Gamma; A], 1)^* \text{Prop}$ and s is the unique section of $p(\mathcal{M}[\Gamma], 1)$ such that

$$\begin{aligned}
 \forall (q(p(\mathcal{M}[\Gamma; A], 1), \text{Prop}) \circ \mathcal{M}[\Gamma, x : A; p]) = \\
 q(p(\mathcal{M}[\Gamma; A], 1), \text{Prop}) \circ \mathcal{M}[\Gamma, x : A; p]
 \end{aligned}$$

$$\begin{array}{ccc}
\mathcal{M}[\Gamma; A] & \xrightarrow{q(p(\mathcal{M}[\Gamma; A], 1), \text{Prop}) \circ \mathcal{M}[\Gamma, x: A; p]} & \text{Prop} \\
& \searrow \mathcal{M}[\Gamma, x: A; p] & \\
& p(\mathcal{M}[\Gamma; A], 1) * \text{Prop} & \xrightarrow{q(p(\mathcal{M}[\Gamma; A], 1), \text{Prop})} \text{Prop} \\
& \downarrow \text{id}_{\mathcal{M}[\Gamma; A]} & \\
& \mathcal{M}[\Gamma; A] & \xrightarrow{p(\mathcal{M}[\Gamma; A], 1)} 1
\end{array}$$

Theorem 2.11 (Correctness Theorem). *For any doctrine of constructions the following correctness conditions, for $\mathcal{M}$ are satisfied:*

1. if Γ ok is derivable, then $\mathcal{M}[\Gamma]$ is defined;
2. if $\Gamma \vdash A$ type is derivable, then $\mathcal{M}[\Gamma; A]$ is defined;
3. if $\Gamma \vdash t: A$ is derivable, then $\mathcal{M}[\Gamma; A]$ and $\mathcal{M}[\Gamma; t]$ are both defined and $\mathcal{M}[\Gamma; t]$ is a section of $\mathcal{M}[\Gamma; A]$;
4. if $\Gamma = \Delta$ is derivable, then $\mathcal{M}[\Gamma]$ and $\mathcal{M}[\Delta]$ are defined and $\mathcal{M}[\Gamma] = \mathcal{M}[\Delta]$;
5. if $\Gamma \vdash A = B$ is derivable, then $\mathcal{M}[\Gamma; A]$ and $\mathcal{M}[\Gamma; B]$ are both defined and $\mathcal{M}[\Gamma; A] = \mathcal{M}[\Gamma; B]$;
6. if $\Gamma \vdash t = s \in A$ is derivable, then $\mathcal{M}[\Gamma; A]$, $\mathcal{M}[\Gamma; t]$ and $\mathcal{M}[\Gamma; s]$ are defined, both $\mathcal{M}[\Gamma; t]$ and $\mathcal{M}[\Gamma; s]$ are sections of $\mathcal{M}[\Gamma; A]$ and $\mathcal{M}[\Gamma; t] = \mathcal{M}[\Gamma; s]$.

Hence, doctrines of constructions are models of Calculus of Constructions.

2.4 The Term Model of Calculus of Constructions

In this section we construct a doctrine of constructions $\mathcal{C}_I$ as a *term model* of calculus of constructions such that the interpretation in this model interprets provably well-defined contexts, types and objects, and identifies only those contexts, types or objects which are provably equal up to renaming of the variables bound in contexts.

In more technical terms, this means that for the interpretation function $\mathcal{M}$ assigning meaning to calculus of constructions in the doctrine of constructions $\mathcal{C}_I$, the following requirements must be fulfilled:

1. for all pre-contexts Γ and Δ , if $\mathcal{M}[\Gamma] = \mathcal{M}[\Delta]$, then $\Gamma = \Delta$ is derivable;
2. for all pre-contexts Γ and Δ and pre-type-expressions A and B , if $\mathcal{M}[\Gamma; A] = \mathcal{M}[\Delta; B]$, then $\Gamma = \Delta$ and $\Gamma \vdash A = B$ are derivable;

3. for all pre-contexts Γ and Δ , pre-type-expressions A and B and pre-object-expressions t and s , if $\mathcal{M}[\Gamma; t] = \mathcal{M}[\Delta; s]$ is a section of $\mathcal{M}[\Gamma; A] = [\Delta; B]$, then $\Gamma = \Delta$, $\Gamma \vdash A = B$ and $\Gamma \vdash t = s : A$ are derivable.

In order to be able to identify contexts up to α -conversion, we consider a fixed numerable set of variables: $v_1, \dots, v_n, \dots$. We say that a context $\Gamma \equiv x_1 : A_1, \dots, x_n : A_n$ is in **α -normal form** iff $x_i \equiv v_i$ for all $1 \leq i \leq n$. Moreover, we say that a pre-context Γ is admissible iff it is in α -normal form and $\vdash \Gamma$ ok.

Now we define $\mathcal{C}_I$: The objects of $\mathcal{C}_I$ are the admissible contexts modulo provable equivalence, i.e., that we will consider two contexts Γ and Δ equal iff $\Gamma = \Delta$ is derivable. By induction over the structure of derivations, one can see that $\Gamma = \Delta$ implies that Γ and Δ have the same length as pre-contexts, so we can define the level of an object of $\mathcal{C}_I$ as the length of any of its representatives. We will write $[\Gamma]$ to represent the class containing Γ . If $\Gamma \equiv v_1 : A_1, \dots, v_n : A_n$ is an admissible context, then we define

$$\text{father}[\Gamma] = [v_1 : A_1, \dots, v_{n-1} : A_{n-1}],$$

that is well defined according to rule CREFL1

If $\Gamma \equiv v_1 : A_1, \dots, v_n : A_n$ and $\Delta \equiv v_1 : B_1, \dots, v_m : B_m$ are admissible contexts, a morphism from $[\Gamma]$ to $[\Delta]$ is given by a m -tuple $\langle t_1, \dots, t_m \rangle$ of terms such that for $i, 1 \leq i \leq m$,

$$\Gamma \vdash t_i : B_i[v_1 \mid t_1, \dots, v_{i-1} \mid t_{i-1}]$$

is derivable, if $\langle s_1, \dots, s_m \rangle$ is another m -tuple such that for $i, 1 \leq i \leq m$

$$\Gamma \vdash s_i : B_i[v_1 \mid s_1, \dots, v_{i-1} \mid s_{i-1}]$$

is derivable, then $\langle t_1, \dots, t_m \rangle$ and $\langle s_1, \dots, s_m \rangle$ are considered equal iff for $i, 1 \leq i \leq m$

$$\Gamma \vdash t_i = s_i : B_i[v_1 \mid s_1, \dots, v_{i-1} \mid s_{i-1}]$$

is derivable, according to rules CREFL1 and CREFL2 the definition is independent from the choice of representatives. We write $[\langle t_1, \dots, t_m \rangle] : [\Gamma] \rightarrow [\Delta]$ for the class of m -tuples equatable to $\langle t_1, \dots, t_m \rangle$. If we consider $\Theta \equiv v_1 : C_1, \dots, v_k : C_k$ is another admissible context and $[\langle s_1, \dots, s_k \rangle] : [\Delta] \rightarrow [\Theta]$, then the composition of $[\langle t_1, \dots, t_m \rangle]$ with $[\langle s_1, \dots, s_k \rangle]$ is defined as

$$[\langle s_1, \dots, s_k \rangle] \circ [\langle t_1, \dots, t_m \rangle] = [\langle s_1[v_1 \mid t_1, \dots, v_m \mid t_m], \dots, s_k[v_1 \mid t_1, \dots, v_m \mid t_m] \rangle].$$

If $\Gamma \equiv v_1 : A_1, \dots, v_n : A_n$ is an admissible context with $n \geq 1$ then the *canonical projection* map $p([\Gamma]) : [\Gamma] \rightarrow \text{father}[\Gamma]$ is represented by the tuple $\langle v_1, \dots, v_{n-1} \rangle$. Let $\Gamma \equiv v_1 : A_1, \dots, v_n : A_n$ and $\Delta \equiv v_1 : B_1, \dots, v_m : B_m$ admissible contexts and

$t = [\langle t_1, \dots, t_m \rangle]: [\Gamma] \rightarrow [\Delta]$ and C a pre-well-formed type expression with $\Delta \vdash C$ type, i.e. $\Delta, v_{m+1}: C$ derivable is an admissible context, then

$$t^*[\Delta, v_{m+1}: C] = [\Delta, v_{m+1}: C[v_1 \mid t_1, \dots, v_m \mid t_m]]$$

$$q([t], [\Delta, v_{m+1}: C]) = [\langle t_1, \dots, t_m, v_{m+1} \rangle: t^*[\Delta, v_{m+1}: C] \rightarrow [\Delta, v_{m+1}: C]].$$

If $\Gamma \equiv v_1: A_1, \dots, v_n: A_n, v_{n+1}: B, v_{n+2}: C$ is an admissible context we define

$$\prod([\Gamma]) = [\Gamma, v_{n+1}: (\prod v_{n+1}: B)C]$$

$$ev_{[\Gamma]} = [\langle v_1, \dots, v_n, v_{n+1}, \text{App}([v_{n+1}: B]C, v_{n+2}, v_{n+1}) \rangle].$$

which can be seen to be a morphism from $[\Gamma, v_{n+1}: B, v_{n+2}: (\prod v_{n+1} B)C]$ to $[\Gamma]$ over the object $[v_1: A_1, \dots, v_n: A_n]$. The object Prop of C_I is defined as $[v_1: \text{Prop}]$ and the object T of C_I is defined as the object $[v_1: \text{Prop}, v_2: \text{Proof}(v_1)]$. If $\Gamma \equiv v_1: A_1, \dots, v_n: A_n, v_{n+1}: B$ and $[\langle p \rangle]$ is a morphism from $[\Gamma]$ to $[v_1: \text{Prop}]$ then we define

$$\forall([p]) = [\langle (\forall v_{n+1}: B)p \rangle]: [v_1: A_1, \dots, v_n: A_n] \rightarrow [v_1: \text{Prop}].$$

It is proven in [11] that C_I is in fact a doctrine of constructions. Also, it is immediate to see that C_I does not interpret or identify more objects or terms than those that are provably defined as equal.

Theorem 2.12 (Completeness Theorem). *Let $\mathcal{M}$ be the interpretation of calculus of construction in C_I . Then*

1. *if $\mathcal{M}[\Gamma]$ is defined then Γ ok is derivable;*
2. *if $\mathcal{M}[\Gamma; A]$ is defined then $\Gamma \vdash A$ type is derivable;*
3. *if $\mathcal{M}[\Gamma; t]$ is defined then $\Gamma \vdash t: A$ is derivable for some pre-type-expression A ;*
4. *if $\Gamma \equiv x_1: A_1, \dots, x_n: A_n, \Delta \equiv y_1: B_1, \dots, y_m: B_m$ and $\mathcal{M}[\Gamma; A] = \mathcal{M}[\Delta; B]$, then $n = m$ and*

$$\begin{aligned} v_1: A_1[x_1 \mid v_1, \dots, x_n \mid v_n], \dots, v_n: A_n[x_1 \mid v_1, \dots, x_n \mid v_n] = \\ = v_1: B_1[y_1 \mid v_1, \dots, y_m \mid v_m], \dots, v_m: B_m[y_1 \mid v_1, \dots, y_m \mid v_m] \end{aligned}$$

$$\begin{aligned} v_1: A_1[x_1 \mid v_1, \dots, x_n \mid v_n], \dots, v_n: A_n[x_1 \mid v_1, \dots, x_n \mid v_n] \\ \vdash A[x_1 \mid v_1, \dots, x_n \mid v_n] = B[y_1 \mid v_1, \dots, y_m \mid v_m] \end{aligned}$$

are derivable;

5. if $\Gamma \equiv x_1 : A_1, \dots, x_n : A_n, \Delta \equiv y_1 : B_1, \dots, y_m : B_m$ and $\mathcal{M}[\Gamma; t] = \mathcal{M}[\Delta, s]$, then $n = m$ and

$$\begin{aligned} v_1 : A_1[x_1 \mid v_1, \dots, x_n \mid v_n], \dots, v_n : A_n[x_1 \mid v_1, \dots, x_n \mid v_n] = \\ = v_1 : B_1[y_1 \mid v_1, \dots, y_m \mid v_m], \dots, v_m : B_m[y_1 \mid v_1, \dots, y_m \mid v_m] \end{aligned}$$

is derivable, and, for some pre-type-expression A ,

$$\begin{aligned} v_1 : A_1[x_1 \mid v_1, \dots, x_n \mid v_n], \dots, v_n : A_n[x_1 \mid v_1, \dots, x_n \mid v_n] \\ \vdash t[x_1 \mid v_1, \dots, x_n \mid v_n] = s[y_1 \mid v_1, \dots, y_m \mid v_m] : A \end{aligned}$$

is derivable.

It can be also shown that $\mathcal{C}_I$ is the initial object of the category of doctrines of constructions and structure preserving functors, named **DoC**. Hence the term model satisfies the properties of soundness, completeness and initiality that we wanted.

2.5 Extending Calculus of Constructions and its Interpretation

In this section we show how to add symbols to Calculus of Constructions, the same thing that we could do in GATs. We need to understand the current CoC as just as a minimal starting point, same as the empty theory for GATs. Inspired by the constructions for Categories with Families presented in [1], we inductively define the concept of signature Σ (corresponding to theory when talking about GATs), CoC extended by Σ , the corresponding category **DoC** $_{\Sigma}$ and interpretation functions $\mathcal{M}^{\Sigma}$.

As a starting case, we consider the empty signature $\Sigma = \emptyset$, together with **CoC** $_{\emptyset}$ representing the current syntax of Calculus of Constructions. The category **DoC** $_{\emptyset}$ is exactly **DoC**, and the notion of interpretation morphism stays the same.

Now, for the inductive step, assume we have defined Σ as a valid signature, and the associated syntax **CoC** $_{\Sigma}$, category **DoC** $_{\Sigma}$ and interpretation function $\mathcal{M}$. Then

- **Adding a sort symbol:** Let Γ be a context in **CoC** $_{\Sigma}$. We can extend Σ with a sort symbol S relative to Γ , to obtain the signature $\Sigma' = (\Sigma, (\Gamma, S))$. Then, we extend **CoC** $_{\Sigma}$ to obtain **CoC** $_{\Sigma'}$ by adding a production of pre-type-expressions $E ::= S$ and the inference rule $\Gamma \vdash S$. The category **DoC** $_{\Sigma'}$ has as objects pairs $(\mathcal{C}, S_{\mathcal{C}})$, where $\mathcal{C}$ is an object of **DoC** $_{\Sigma}$ and $\mathcal{M}_{\mathcal{C}}[\Gamma] \triangleleft A_{\mathcal{C}}$. Finally, we define $\mathcal{M}'_{\mathcal{C}}$, for each object $\mathcal{C}$ of **DoC** $_{\Sigma'}$, by adding $\mathcal{M}'_{\mathcal{C}}[\Gamma; S] = S_{\mathcal{C}}$.

- **Adding an operator symbol:** Let Γ be a context and A be a type under Γ of CoC_Σ , we can extend Σ with a new operator symbol f relative to Γ and A obtaining $\Sigma' = (\Sigma, (\Gamma, A, f))$. Then, we extend CoC_Σ to obtain $\text{CoC}_{\Sigma'}$ by adding a production of pre-object-expressions $t ::= f$ and the inference rule $\Gamma \vdash f: A$. The category $\mathbf{DoC}_{\Sigma'}$ has as objects pairs $(\mathcal{C}, f_{\mathcal{C}})$, where $\mathcal{C}$ is an object of $\mathbf{DoC}_\Sigma$ and $f_{\mathcal{C}}$ is a section of $\mathcal{M}_{\mathcal{C}}[\Gamma, x: A]$. Finally, we define $\mathcal{M}'_{\mathcal{C}}$, for each object $\mathcal{C}$ of $\mathbf{DoC}_{\Sigma'}$, by adding $\mathcal{M}'_{\mathcal{C}}[\Gamma; f] = f_{\mathcal{C}}$.
- **Adding an equality of types:** Let Γ be a context, A, A' types under Γ of CoC_Σ , we can extend Σ with a new equation $A = A'$ relative to Γ , to obtain $\Sigma' = (\Sigma, (\Gamma, A, A'))$. Then, we extend CoC_Σ to obtain $\text{CoC}_{\Sigma'}$ by adding the inference rule $\Gamma \vdash A = A'$. The category $\mathbf{DoC}_{\Sigma'}$ is a full subcategory of $\mathbf{DoC}_\Sigma$ containing the objects $\mathcal{C}$ of $\mathbf{DoC}_\Sigma$ such that $\mathcal{M}_{\mathcal{C}}[\Gamma; A] = \mathcal{M}_{\mathcal{C}}[\Gamma; A']$. $\mathcal{M}'$ is the restriction of $\mathcal{M}$ to the objects of $\mathbf{DoC}_{\Sigma'}$.
- **Adding an equality of terms:** Let Γ be a context, A a type under Γ and a, a' terms of type A under Γ of CoC_Σ . We can extend Σ with a new equation $a = a'$ relative to Γ and A , to obtain $\Sigma' = (\Sigma, (\Gamma, A, a, a'))$. Then, we extend CoC_Σ to obtain $\text{CoC}_{\Sigma'}$ by adding the inference rule $\Gamma \vdash a = a': A$. The category $\mathbf{DoC}_{\Sigma'}$ is a full subcategory of $\mathbf{DoC}_\Sigma$ containing the objects $\mathcal{C}$ of $\mathbf{DoC}_\Sigma$ such that $\mathcal{M}_{\mathcal{C}}[\Gamma; a] = \mathcal{M}_{\mathcal{C}}[\Gamma; a']$. $\mathcal{M}'$ is the restriction of $\mathcal{M}$ to the objects of $\mathbf{DoC}_{\Sigma'}$.

Under this construction, we can consider $\mathcal{C}_\Sigma$ to be the exact same construction as $\mathcal{C}_I$, but with CoC_Σ instead of $\text{CoC}_\emptyset$. Following a similar proof that the original proof for $\mathcal{C}_I$ in [11] we can prove the corresponding *Completeness Theorems* and that $\mathcal{C}_\Sigma$ is the initial object of $\mathbf{DoC}_\Sigma$.

2.6 Redefinition of the Problem

Now, we want to redefine our problem from the previous definition for GATs to our new syntax, then rephrase it to similar categorical terms such that an extended version of the expansion algorithm works.

The problem now can be formalized as follows: Given a valid signature of Calculus of Constructions Σ and a derivable sequent $\Gamma \vdash A$ type in CoC_Σ , we want to find a pre-object-expression t such that $\Gamma \vdash t: A$ is a derivable rule in CoC_Σ .

We can consider this from the point of view of the category $\mathcal{C}_\Sigma$ introduced in the previous section. This means that our problem is equivalent to finding a section s of $\mathcal{M}[\Gamma; A] = [\Gamma, v_{n+1}: A]$ in $\mathcal{C}_\Sigma$, assuming Γ is a context of length n in normal form.

Moreover, we want to use the reduction of our search space we did on the previous case by considering the category over a base object. To do it we need the following result:

Definition 2.13. Given a doctrine of constructions $\mathcal{C}$, and an arbitrary object A of $\mathcal{C}$. The *relativization* of $\mathcal{C}$ to A , noted as $\mathcal{C}_A$, is defined as follows:

1. $\mathcal{C}_A$ is a category whose objects are the collection of objects B over A in $\mathcal{C}$, i.e. objects B in $\mathcal{C}$ such that $A \leq B$;
2. for two objects B, C in $\mathcal{C}_A$, a morphism from B to C over A , i.e., a morphism in $\mathcal{C}_A$, is a morphism $f: B \rightarrow C$ in $\mathcal{C}$ such that $p(C, A) \circ f = p(B, A)$, composition of morphisms is inherited from $\mathcal{C}$;
3. $\text{father}_A(B) = \text{father}(B)$ for $B > A$ and $\text{father}_A(A) = \text{father}(A)$;
4. for objects $B > A$, we define $p_A(B) = p(B): B \rightarrow \text{father}_A(B)$;
5. for B, C, D objects of $\mathcal{C}$ with $\text{father}_A(D) = C$ and morphism $f: B \rightarrow C$ in $\mathcal{C}_A$ we define

$$f^{*A}D = f^*D \quad \text{and} \quad q_A(f, D) = q(f, D);$$

6. for B in $\mathcal{C}$ with $\text{level}_A(B) \geq 2$,

$$\prod_A(B) = \prod(B) \quad \text{and} \quad \text{eval}_B^A = \text{eval}_B;$$

7. if T is the collection of types of $\mathcal{C}$, then

$$T_A = p(A)^*T$$

is the collection of types of $\mathcal{C}_A$.

Theorem 2.14. The relativization $\mathcal{C}_A$ of a doctrine of constructions $\mathcal{C}$ to an object A is a doctrine of constructions.

Proof.

□

With this result, and with the notation of Γ subscript introduced in the second chapter, we reduce our problem to finding a section of $[\langle x: A \rangle_\Gamma]$ in $\mathcal{C}_{\Sigma, \Gamma}$, the relativization of $\mathcal{C}_\Sigma$ to Γ .

2.7 Expansion Algorithm

Now we extend the expansion algorithm from the previous chapter. We reintroduce the rules in order to match the syntax we are using for Calculus of Constructions. First, consider G to be a finite subcategory of $\mathcal{C}_{\Sigma, \Gamma}$ containing $[\Gamma]$, $[\Gamma, p: \text{Prop}]$, $\Gamma, x: A$ and their projections. Now we define the expansion rules. If Δ is a context, then:

- E1 For each sort symbol in the signature Σ , with an introductory rule $\Omega \vdash S$ type, such that $\Delta \equiv \Gamma, y_1: B_1, \dots, y_m: B_m$, $\Omega \equiv z_1: C_1, \dots, z_w: C_w$ and G_n contains sections

$$\begin{aligned} [\Delta] &\xrightarrow{\langle y_1, \dots, y_m, t_1 \rangle_\Gamma} [\Delta, z_1: C_1] \\ [\Delta] &\xrightarrow{\langle y_1, \dots, y_m, t_2 \rangle_\Gamma} [\Delta, z_2: C_2[z_1 \mid t_1]] \\ &\vdots \\ [\Delta] &\xrightarrow{\langle y_1, \dots, y_m, t_w \rangle_\Gamma} [\Delta, z_w: C_w[z_1 \mid t_1, \dots, z_{w-1} \mid t_{w-1}]]. \end{aligned}$$

Then we add $[\Delta, y: A[z_1 \mid t_1, \dots, z_w \mid t_w]]$ and its projection to G_{n+1} .

- E2 If $\Delta \equiv \Gamma, y_1: B_1, \dots, y_m: B_m$, then we add the sections

$$\begin{aligned} [\Delta] &\xrightarrow{\langle y_1, \dots, y_m, y_i \rangle_\Gamma} [\Delta, y: B_i] \\ [\Delta] &\xrightarrow{\langle y_1, \dots, y_m, x_j \rangle_\Gamma} [\Delta, x: A_j] \end{aligned}$$

and their corresponding projections for each $1 \leq i \leq m$ and each $1 \leq j \leq n$.

- E3 For each sort symbol in the signature Σ , with an introductory rule $\Omega \vdash F: A$, such that $\Delta \equiv \Gamma, y_1: B_1, \dots, y_m: B_m$, $\Omega \equiv z_1: C_1, \dots, z_w: C_w$ and the following sections exist in G_n

$$\begin{aligned} [\Delta] &\xrightarrow{\langle y_1, \dots, y_m, t_1 \rangle_\Gamma} [\Delta, z_1: C_1] \\ [\Delta] &\xrightarrow{\langle y_1, \dots, y_m, t_2 \rangle_\Gamma} [\Delta, z_2: C_2[z_1 \mid t_1]] \\ &\vdots \\ [\Delta] &\xrightarrow{\langle y_1, \dots, y_m, t_w \rangle_\Gamma} [\Delta, z_w: C_w[z_1 \mid t_1, \dots, z_{w-1} \mid t_{w-1}]] \end{aligned}$$

we add the section

$$[\Delta] \xrightarrow{\langle y_1, \dots, y_m, F[z_1 \mid t_1, \dots, z_w \mid t_w] \rangle_\Gamma} [\Delta, z: A[z_1 \mid t_1, \dots, z_w \mid t_w]].$$

E4 If $\Delta \equiv \Gamma, y_1 : B_1, \dots, y_m : B_m$, for every morphism

$$[\langle t_1, \dots, t_{m-1} \rangle_\Gamma] : [\Gamma, z_1 : C_1, \dots, z_w : C_w] \rightarrow [\Gamma y_1 : B_1, \dots, y_{m-1} : B_{m-1}]$$

we construct the corresponding pullback:

$$\begin{array}{ccc} \Gamma, z_1 : C_1, \dots, z_w : C_w, y_m : B_m [y_1 \mid t_1, \dots, y_{m-1} \mid t_{m-1}] & \xrightarrow{\langle t_1, \dots, t_{m-1}, y_m \rangle_\Gamma} & \Gamma, y_1 : B_1, \dots, y_m : B_m \\ \downarrow & & \downarrow \\ \Gamma, z_1 : C_1, \dots, z_w : C_w & \xrightarrow{\langle t_1, \dots, t_{m-1} \rangle_\Gamma} & \Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1} \end{array}$$

to G_{n+1} .

E5 If $\Delta \equiv \Gamma, v_1 : B_1, \dots, v_m : B_m, v_{m+1} : C, v_{n+2} : D$ we add

$$\prod([\Delta]) = [\Gamma, v_1 : B_1, \dots, v_m : B_m, v_{m+1} : (\prod v_{m+1} : C)D]$$

$$\text{eval}_{[\Delta]} = [\Gamma, v_1, \dots, v_{m+1}, \text{App}([v_{m+1} : C]D, v_{m+2}, v_{m+1})].$$

a morphism from $[\Gamma, v_1 : A_1, \dots, v_m : B_m, v_{n+1} : C, v_{n+2} : (\prod v_{n+1} : B)C]$ to $[\Delta]$, and the corresponding projections.

E6 If $\Delta \equiv \Gamma, v_1 : B_1, \dots, v_m : B_m, v_{m+1} : C$, for any morphism $[\langle p \rangle_\Gamma] : [\Delta] \rightarrow [\langle v_1 : \text{Prop} \rangle_\Gamma]$ we add

$$\begin{aligned} \forall([\langle p \rangle_\Gamma]) &= [\langle (\forall v_{m+1} C) p \rangle_\Gamma] : [\Gamma, v_1 : B_1, \dots, v_m : B_m] \rightarrow [\langle v_1 : \text{Prop} \rangle_\Gamma] \\ &[\Delta, x : \text{Proof}(p)] \end{aligned}$$

and its corresponding projection.

After each expansion, we complete the category in the following way: for every product object, i.e., for each object $[\Delta]$ with Δ of the form $\Omega, (\prod x : A)B$, and every section $s = \langle x, a \rangle_\Omega$ of $[\Omega, x : A, y : B]$ we add $\text{curry}(s) = \langle (\lambda x : A) a \rangle_\Omega$ as a section of $[\Delta]$. Also, we add the composition of morphisms to G_{n+1} in order to complete it as a subcategory of $\mathcal{C}_{\Sigma, \Gamma}$.

Theorem 2.15. *Every object and every morphism of $\mathcal{C}_{\Sigma, \Gamma}$ are constructible by expansion.*

Proof. We prove this result by induction over the level of contexts, pre-type-expressions and terms.

Consider a context $\Delta \equiv \Gamma, y_1 : B_1, \dots, y_m : B_m$ in CoC_Σ , and the corresponding object $[\Delta]$ in $\mathcal{C}_{\Sigma, \Gamma}$. We prove that this object is constructible by expansion. We consider how the sequent $\Delta \text{ ok}$ was derived. It has to be by one of the following rules: **EMPTY**, **CONT-INTRO**, **CONT-THIN**, **CONT-SUB**, **CREFL** and **REFLECTION1**. Since the case of **CREFL** and **REFLECTION1** are trivial, we consider each of the other cases:

EMPTY This is the base case, when Δ is the empty context. This may only happen when $\Gamma = \Delta = \langle \rangle$ and $m = 0$, and as the object $[\Gamma]$ is included in G , it is always constructible by expansion.

CONT-INTRO This means that the sequent $\Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1} \vdash A_m$ type is derivable. Which itself can only be derived using the following rules: **CONT-INTRO** (in the reverse form), **REFL-TYP**, **[]-FORM**, **PROP**, **PROP-TYPE** and any introductory rule of a sort symbol in Σ . The derivation from rules **CONT-INTRO** and **REFL-TYP** are trivial, as they both enforce the existence of the object $[\Delta]$ by premise, hence we will consider the other cases:

[]-FORM Then, we have that A_m is a pre-type-expression of the form $([] x : N)M$ for some variable x and two pre-type-expressions N, M such that the sequent $\Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}, x : N \vdash M$ type is derivable. By induction hypothesis, the object

$$\begin{aligned} \mathcal{M}[\Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}, x : N; M] &= \\ &= [\Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}, x : N, y : M] \end{aligned}$$

is constructible by expansion. By the expansion rule E5 applied to that object, gives us that

$$\begin{aligned} []([\Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}, x : N, y : M]) &= \\ &= [\Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}, x : ([] x : N)M] \end{aligned}$$

is constructible, that is equal to $[\Delta]$ as they just differ by the naming of bound variables. Hence, $[\Delta]$ is constructible by expansion.

PROP Then we have that A_m is equal to **Prop** as a pre-type-expression, and and, with $\Delta' \equiv \Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}$, Δ' ok is a derivable sequent. By induction hypothesis, we know that $[\Delta']$ is constructible by expansion. Also, as G contains $[\Gamma, x : \text{Prop}]$ and it's projection to $[\Gamma]$, by applying the expansion rule E4 to $[\Gamma, x : \text{Prop}]$ with the projection morphism from $[\Delta']$ to $[\Gamma]$, we get that the following pullback diagram is constructible by expansion:

$$\begin{array}{ccc} [\Delta', x : \text{Prop}] & \xrightarrow{\langle x \rangle_\Gamma} & [\Gamma, x : \text{Prop}] \\ \downarrow & & \downarrow \\ [\Delta'] & \longrightarrow & [\Gamma] \end{array}$$

and $[\Delta', x : \text{Prop}] = [\Delta]$ as they only differ by the naming of the variables.

PROP-TYPE Then we have that there is a pre-object-expression p such that A_m equals $\text{Proof}(p)$ as a pre-type-expression and the sequent $\Delta' \vdash p : \text{Prop}$, with $\Delta' \equiv \Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}$, is derivable. By induction hypothesis, $[\Delta']$ is constructible by expansion. Moreover, by applying induction hypothesis,

$$\mathcal{M}[\Delta'; p] = \langle p \rangle_\Gamma$$

is constructible by expansion. Finally, by applying expansion rule E6, we get that $[\Delta] = [\Delta', x : \text{Proof}(p)]$ is constructible by expansion.

INTRODUCTORY RULE Consider a sort symbol S with the corresponding introductory rule $\Omega \vdash S$, with $\Omega \equiv z_1 : C_1, \dots, z_k : C_k$, and $t_1, \dots, t_k$ pre-object-symbols such that $A_m = S[z_1 \mid t_1, \dots, z_k \mid t_k]$. Then, for each i , $1 \leq i \leq k$, $\Delta' \vdash t_i : C_i[z_1 \mid t_1, \dots, z_{i-1} \mid t_{i-1}]$ are derivable judgments, where $\Delta' \equiv \Gamma, y_1 : B_1, \dots, y_{m-1} : B_{m-1}$. Hence, by inductive hypothesis, the sections

$$\mathcal{M}[\Delta'; t_1] : [\Delta'] \rightarrow [\Delta', z_1 : C_1]$$

$$\mathcal{M}[\Delta'; t_2] : [\Delta'] \rightarrow [\Delta', z_2 : C_2[z_1 \mid t_1]]$$

$$\vdots$$

$$\mathcal{M}[\Delta'; t_k] : [\Delta'] \rightarrow [\Delta', z_k : C_k[z_1 \mid t_1, \dots, z_{k-1} \mid t_{k-1}]]$$

are constructible by expansion. Then, by expansion rule E1, $[\Delta] = [\Delta', y : S[z_1 \mid t_1, \dots, z_m \mid t_m]]$ is constructible by expansion.

CONT-THIN Since context Δ is of the form $\Gamma, y_1 : B_1, \dots, y_m : B_m$, then there exists an k , $1 \leq k \leq m$ such that $\Gamma, y_1 : B_1, \dots, y_{k-1} : B_{k-1}, y_{k+1} : B_{k+1}, \dots, y_m : B_m$ is a context and $\Gamma, y_1 : B_1, \dots, y_{k-1} : B_{k-1} \vdash B_k$ type is a derivable sequent. By induction hypothesis, if $\Delta' \equiv \Gamma, y_1 : B_1, \dots, y_{k-1} : B_{k-1}$, $[\Delta']$ is constructible by expansion, and by the same argument in the last point, so is $[\Delta', y_k : B_k]$. Finally, by considering the expansion rule E4 on $[\Delta', y_{k+1} : B_{k+1}, \dots, y_m : B_m]$ and the projection morphism of $[\Delta', y_k : B_k]$ onto $[\Delta']$, we get that the pullback diagram

$$\begin{array}{ccc} \langle \Delta' \rangle_{\Gamma}^{y_1, \dots, y_{k-1}, y_{k+1}, \dots, y_m} & \xrightarrow{\quad} & [\Delta', y_{k+1} : B_{k+1}, \dots, y_m : B_m] \\ \downarrow & & \downarrow \\ [\Delta', y_k : B_k] & \longrightarrow & [\Delta'] \end{array}$$

is constructible by expansion, hence, so is $[\Delta]$.

CONT-SUB Then there exists a k , $1 \leq k \leq m$, a variable x not appearing in Δ , pre-type-expressions $A, B'_k, \dots, B'_m$ and a pre-object-expression t such that $\Gamma, y_1 : B_1, \dots, y_{k-1} : B_{k-1}, x : A, y_k :$ and $\Gamma, y_1 : B_1, \dots, y_{k-1} : B_{k-1} \vdash t : A$ are derivable sequents and $B_i = B'_i[x \mid t]$ for $k \leq i \leq m$. For simplicity of notation, we will note $\Delta' \equiv \Gamma, y_1 : B_1, \dots, y_{k-1} : B_{k-1}$. Then, by induction hypothesis we have that both contexts $[\Delta', x : A]$ and $[\Delta', x : A, y_k : B'_k, \dots, y_m : B'_m]$ and the section $[\langle y_1, \dots, y_{k-1}, t \rangle_\Gamma]$ are constructible by expansion. Hence, by rule E4, so is the following pullback diagram:

$$\begin{array}{ccc} [\Delta', y_k : B'_k[x \mid t], \dots, y_m : B'_m[x \mid t]] & \xrightarrow{[\langle s, y_k, \dots, y_m \rangle_{\Delta'}]} & [\Delta', x : A, y_k : B'_k, \dots, y_m : B'_m] \\ \downarrow & & \downarrow \\ [\Delta'] & \xrightarrow{[\langle s \rangle_{\Delta'}]} & [\Delta', x : A] \end{array}$$

Finally, $[\Delta] = [\Delta', y_k : B'_k[x \mid t], \dots, y_m : B'_m[x \mid t]]$ is constructible by expansion.

Now consider a morphism

$$[\langle t_1, \dots, t_w \rangle_\Gamma] : [\Gamma, y_1 : B_1, \dots, y_m : B_m] \rightarrow [\Gamma, z_1 : C_1, \dots, z_w : C_w]$$

in $\mathcal{C}_{\Sigma, \Gamma}$. We denote $\Delta \equiv \Gamma, y_1 : B_1, \dots, y_m : B_m$. Then for i , $1 \leq i \leq w$,

$$\Delta \vdash t_i : C_i[z_1 \mid t_1, \dots, z_{i-1} \mid t_{i-1}]$$

are derivable sequents of CoC_Σ . Now we want to see that, for each i , the following section is constructible by expansion:

$$[\langle y_1, \dots, y_m, t_i \rangle_\Gamma] : [\Delta] \rightarrow [\Delta, z_i : C_i[z_1 \mid t_1, \dots, z_{i-1} \mid t_{i-1}]].$$

We consider the ways the mentioned sequents are derived. They must be derived by one of the following rules: ASS, REFL-OBJ, CONV1, $\llbracket \cdot \rrbracket$ -INTRO, $\llbracket \cdot \rrbracket$ -ELIM, PROP-TYPE and $\forall$ -INTRO. The cases of REFL-OBJ and CONV1 are trivial. Hence, we consider the other cases:

Ass There exists an $1 \leq j \leq m$ or $1 \leq k \leq n$ such that

$$t_i = y_j \quad \text{orange} \quad t_i = x_k.$$

We suppose that $t_i = y_j$ for some j , and the other case is equivalent. In that case $B_j = C_i[z_1 \mid t_1, \dots, z_{i-1} \mid t_{i-1}]$. We know that Δ is a context, hence by inductive hypothesis we have that $[\Delta]$ is constructible by expansion. When applying the expansion rule E2, we obtain that the section

$$[\langle y_1, \dots, y_m, y_j \rangle_\Gamma] : [\Delta] \rightarrow [\Delta, y : B_j]$$

is constructible by expansion, and it is equal to $[\langle y_1, \dots, y_m, t_i \rangle_\Gamma]$, hence the least is also constructible by expansion.

Π -INTRO Then there exists one pre-object-expression t and two pre-type-expression A, B such that $t_i = (\lambda x: A)t$ and $C_i[z_1 \mid t_1, \dots, z_{i-1} \mid t_{i-1}] = (\Pi x: A)B$, for some variable x . Moreover, as $\text{level}[\Delta; (\lambda x: A)t] = \text{level}[\Delta, x: A; t] + 1$, by induction hypothesis, the section

$$[\langle y_1, \dots, y_m, x, t \rangle_\Gamma]: [\Delta, x: A] \rightarrow [\Delta, x: A, y: B]$$

is constructible by expansion. Applying the expansion rule E5 to $[\Delta, x: A, y: B]$, we get that $[\Delta, f: (\Pi x: A)B]$ is constructible by expansion. Finally, by the curry completion condition, we get that

$$\text{curry}(\langle y_1, \dots, y_m, x, t \rangle_\Gamma) = \langle y_1, \dots, y_m, (\lambda x: A)t \rangle_\Gamma$$

is constructible by expansion.

Π -ELIM Then there exists one variable x , two pre-type-expressions A, B and two pre-term-expressions t, s such that $\Delta' \vdash t: (\Pi x: A)B$, $\Delta' \vdash s: A$, $\Delta', x: A \vdash B$ type and $B_m = \text{App}([x: A]B, t, s)$. TODO: finish this, photo of partial proof

PROP-TYPE TODO

$\forall$ -INTRO TODO

□

Chapter 3

Implementation of the Expansion Algorithm

In this chapter we explain the basic ideas of how to use the expansion algorithm jointly with graph neural networks with the objective of generating terms of a specific type in a given theory. We also present a demo of the raw expansion algorithm for generating terms on simply typed combinatory logic in order to give some intuition on how the algorithm works.

3.1 Term Generation in Simply Typed Combinatory Logic

Now we present the `IMPLICA` package for Python [7], a graph engine built on Rust that we built for the purpose of this demonstration. The idea of this package is to allocate the finite subcategory G , and help to apply the different expansion iterations via its cypher-like interface directly from Python.

We start by stating what simply typed Combinatory Logic, or $\mathbf{CL}_{\rightarrow}$, is. Given a countable set of type variables $P_0, P_1, P_2, \dots$, we define the set of *simple types* $\mathcal{T}_{\rightarrow}$ as the set generated by the following clauses:

1. type variables belong to $\mathcal{T}_{\rightarrow}$;
2. if $A, B \in \mathcal{T}_{\rightarrow}$, then $(A \rightarrow B) \in \mathcal{T}_{\rightarrow}$.

Types of the form $A \rightarrow B$ are called *function types*, and are the simply typed version of the product types we introduced in CoC. Terms of $\mathbf{CL}_{\rightarrow}$ must have a type assigned and are generated by the following clauses:

1. for each $A \in \mathcal{T}_{\rightarrow}$ there is a countably infinite supply of variables of type A ;

2. if t, s are terms of types $A \rightarrow B$ and A respectively, then $\text{App}(t, s)$ is a term of type B ;
3. for all $A, B, C \in \mathcal{T}_{\rightarrow}$ there are constant terms

$$\mathbf{k}^{A,B} : A \rightarrow B \rightarrow A$$

$$\mathbf{s}^{A,B,C} : (A \rightarrow B \rightarrow C) \rightarrow (A \rightarrow B) \rightarrow A \rightarrow C.$$

We use the following notation: To assert that t is a term of type A , we write $t : A$; instead of writing $\text{App}(t, s)$ we can simply write ts .

We can write this as a GAT signature as follows:

$$\begin{aligned} & \vdash \text{Prop type} \\ & p : \text{Prop} \vdash \text{Proof}(p) \text{ type} \\ & A, B : \text{Prop} \vdash (A \rightarrow B) : \text{Prop} \\ & A, B : \text{Prop}, t : \text{Proof}(A \rightarrow B), s : \text{Proof}(A) \vdash \text{App}(t, s) : \text{Proof}(B) \\ & A, B : \text{Prop} \vdash \mathbf{k}^{A,B} : \text{Proof}(A \rightarrow B \rightarrow A) \\ & \text{begin align} * A, B, C : \text{Prop} \vdash \mathbf{s}^{A,B,C} : \text{Proof}((A \rightarrow B \rightarrow C) \rightarrow (A \rightarrow B) \rightarrow A \rightarrow C) \end{aligned}$$

This allows us to leverage the construction of $C(U)$ we made for the general case, but this will be rather difficult to follow. Hence, we will look for simplification both of the category and the expansion rules. Consider Γ a simple type expression of $\mathbf{CL}_{\rightarrow}$, let $A_1, \dots, A_n$ be the type variables appearing on Γ , and let Δ be the context $A_1 : \text{Prop}, \dots, A_n : \text{Prop}$. We center our search on the relativization of $C(U)$ by Δ , $\mathcal{C}_{\Delta}$. This is a contextual category, with all the tree structure, but this tree structure carries little-to-none meaning as we are working with "non-dependent" types, so by just considering objects of the first level and the final object we have all the possible types in $A_1, \dots, A_n$ and their relevant morphisms. Moreover, the object $[\Delta, x : \text{Prop}]$ lies in the level 1, but it carries no important information in our case, as its just used for object formation by the general expansion algorithm, but that will be done differently in this special case, so we will remove it. So we can consider $\mathcal{C}$ to be the full subcategory of $C(U)$ containing $[\Delta]$ and $[\Delta, x : \text{Proof}(p)]$ for each term p of type Prop under context Δ . The objects of this category are in 1-1-correspondence with the elements of $\mathcal{T}_{\rightarrow} \cup \{1\}$, hence we note

$$1 := [\Delta] \quad \text{and} \quad p := [\Delta, x : \text{Proof}(p)].$$

In the case of morphisms, we will note them by the term appearing in the $n + 1$ component, i.e., each morphism $\langle A_1, \dots, A_n, s \rangle$ in $\mathcal{C}$ will be noted as just s . With this notation in mind we define the following rules of expansion: Let p be an element of $\mathcal{T}_{\rightarrow}$,

1. if p is of the form $B \rightarrow A$, we add

$$A \xrightarrow{\mathbf{k}^{A,B}} p;$$

2. for each q in G_n we add

$$p \xrightarrow{\mathbf{k}^{p,q}} q \rightarrow p;$$

3. if p is of the form $(B \rightarrow A) \rightarrow A \rightarrow C$, we add

$$(A \rightarrow B \rightarrow C) \xrightarrow{\mathbf{s}^{A,B,C}} p;$$

4. if p is of the form $(A \rightarrow B \rightarrow C)$, we add

$$p \xrightarrow{\mathbf{s}^{A,B,C}} ((A \rightarrow B) \rightarrow A \rightarrow C);$$

Also, when creating an object p , if p is of the form $A \rightarrow B \rightarrow A$, add the morphism $\mathbf{k}^{A,B}: 1 \rightarrow p$, and if p is of the form $(A \rightarrow B \rightarrow C) \rightarrow (A \rightarrow B) \rightarrow A \rightarrow C$, add the morphism $\mathbf{s}^{A,B,C}: 1 \rightarrow p$. Then, complete G_{n+1} with the composition of morphisms in order for it to be a subcategory of $\mathcal{C}$. The exhaustivity of this expansion algorithm is easy to check, as it follows directly from the definitions.

Now we show how to code the expansion algorithm using Python and the `IMPLICA` package. One peculiarity we developed into `IMPLICA` that is not part of our current definition is the fact that all terms of a type are considered equal, as we just care about having a witness, not the nature of it. In order to add this condition to the GAT signature, it is enough to add the following equation:

$$p: \text{Prop}, t_1, t_2: \text{Proof}(p) \vdash t_1 = t_2$$

As it is not really part of combinatory logic, we left it out of its formal definition, but for our task it is more efficient to consider it this way.

First we initialize an `IMPLICA` graph and we add the proposition $A \rightarrow A$ using the query API.

```
import implica

# Initialize the graph with Combinatory Logic constants
graph = implica.Graph(
    constants = [
        implica.Constant(("K", "(A:*)->(B:*)->A")),
        implica.Constant("S", "((A:*)->(B:*)->(C:*))->(A->B)->A->C")
    ]
)
```

```
)

# Seed the graph with our target objective (A -> A)
graph.query().create(f"(:{query})").execute()
```

Now, to perform one expansion iteration we execute the following code:

```
# Mark all current nodes as 'existed' to differentiate them from new generations
graph.query().match("N:*").set("N", { "existed" : True }).execute()

# Rule 1 & 2: K-Combinator Expansions
(
    self.graph.query()
    # Find existing nodes matching the signature of a K-target
    .match("N:(B:*) -> (A:*) { existed: true }")
    # Create the precursor node and the connecting morphism
    .create("(: A { existed: false })-[::@K(A, B)]->(N)")
    .execute()
)
(
    self.graph.query()
    .match("N:(A:*) { existed: true }")
    # CRITICAL: Limit creation number to prevent memory overflow
    .limit(10)
    .match("M:(B:*) { existed: true }")
    .create("(N)-[::@K(A, B)]->(:B -> A { existed: false })")
    .execute()
)
# Rule 3 & 4: S-Combinator Expansions
(
    self.graph.query()
    .match("N:((A:*) -> (B:*)) -> A -> (C:*) { existed: true }")
    .create("(:A->B->C { existed: false })-[::@S(A, B, C)]->(N)")
    .execute()
)
(
    self.graph.query()
    .match("N:(A:*)->(B:*)->(C:*) { existed: true }")
    .create("(N)-[::@S(A, B, C)]->(:A -> B) -> A -> C { existed: false }")
    .execute()
)
```

In the first line, we set an attribute "existed" to true in order to distinguish between new nodes and nodes already present at the beginning. It is important to state the *.limit* modifier on the second query, because this expansion rule creates, or tries to create a new node for each pair of nodes in the graph, hence it makes complexity explode. To solve this we use the modifier *.limit* to cap the number of creations per original node to 10. Feel free to remove this line. This explosive growth of the complexity of the algorithm is what we will try to solve later by adding graph neural networks.

Let us walk through, iteration by iteration, on the generation of a term of type $A \rightarrow A$ by the expansion algorithm to get some intuition. For simplicity, we have omitted the 1 element on the graph displays, and also, as we just want witnesses of the existence of a term, we just have at most one morphism between two objects, and the sections (morphisms from 1) will be displayed simply as red dots instead of blue. The starting point G_0 is rather uninteresting, as it just contains the object $A \rightarrow A$ with no section. In the Figure B.1a, we can see that G_1 contains three unpopulated objects A , $A \rightarrow A$ and $(A \rightarrow A) \rightarrow A \rightarrow A$ from left to right. Then, by applying another expansion, we obtain G_2 , Figure B.1b, where we start to see some objects getting populated, for example $(A \rightarrow A) \rightarrow A \rightarrow A$ with term **sk**. By the next expansion, G_3 grows significantly, see Figure B.1c, but $A \rightarrow A$ remains unpopulated. Finally, in G_4 $A \rightarrow A$ gets inhabited by the term **skk**, but as can be seen in Figure B.1d, the exponential growth of the graph continues.

Iteration (G_n)	Nodes (Capped at 10/match)	Nodes (Uncapped)
G_0	1	1
G_1	3	3
G_2	11	11
G_3	116	123
G_4	1352	15131

Table 3.1: Node count explosion during the expansion algorithm for the proof of $A \rightarrow A$ in $CL_{\rightarrow}$.

Now that we have a clearer understanding of how the expansion algorithm works, we introduce the optimizations for the general case.

3.2 Graph Neural Networks

Graph Neural Networks (GNNs) process complex data structures by leveraging relationships between individual data points (nodes) and their connections

(edges) through a mechanism known as "message passing". During this process, each node iteratively updates its own representation by gathering and aggregating information from its immediate neighbors, enabling the model to learn both the local features of the data and the broader structural geometry of the network. Because of this unique architecture, GNNs are highly effective for node classification (assigning specific labels to individual nodes), graph classification (categorizing the entire network structure as a single entity), and graph embedding (translating complex graph structures into lower-dimensional vectors for easier processing in downstream machine learning tasks).

The idea behind the expansion algorithm comes from trying to use GNNs to take advantage of the graph structures that emerge from the categorical semantics we presented in the previous chapters. So we built it to work together with a GNN performing node classification. First consider the GNN as just a simple labeler, that given the finite subcategory G_n produced by the previous expansion, it labels each node in the following 3 categories: KEEP, EXPAND and DELETE. Then we define a partial expansion phase where we compute G_{n+1} from G_n without those nodes labeled as DELETE, and by only applying expansion on those nodes marked as EXPAND. This change will make us lose the exhaustivity of the process, but we will be gaining a lot of efficiency if the GNN manages to prune all the useless nodes and expand only those nodes we really need for our generation. In the ideal case, i.e. the GNN always labels nodes correctly, we will be able to still generate terms of any type but we would make it with far less complexity.

Although the implementation of the following algorithm is left for further work, we present the basic ideas for building our labeling model. The first thing we must note is that using supervised learning techniques (that need to have input-output pairs) need for us to be able to generate a large quantity of training examples by the execution of the raw expansion algorithm, hence we have to discard this option as we would just be able to train our model on very simple examples, due to the high complexity of the algorithm. To solve this, we would need to train using unsupervised learning. The technique we are going to use is called Teacher-Student or Conjecturer-Prover. It consists of having a two-headed model, i.e., a model with a GNN based body together with two different projection heads. The first head or conjecturer will propose a challenge, or in our case a type we need to find a term of, and the second head or prover will act as the GNN we talked before and try to find the term with the partial expansion algorithm. Then, if a solution is found, we compute the optimal labels at each step to compute the loss of the prover head. Then we compute the loss of the conjecturer by measuring how far it was the prover's loss from some objective loss. This way we make the conjecturer output problems with the right level of complexity.

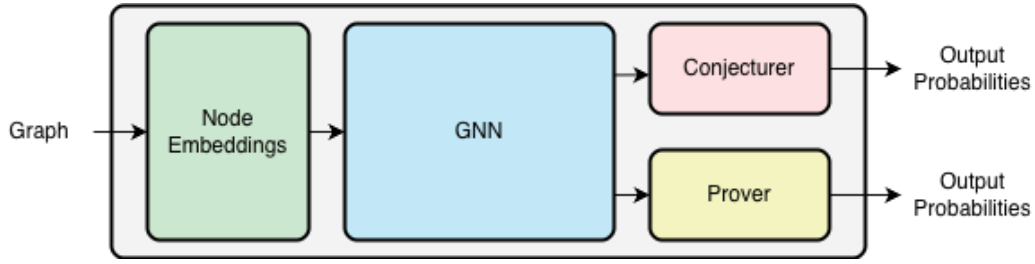


Figure 3.1: Base architecture diagram

We study the components separately:

Node Embedding: The objective of this component is to assign numerical embeddings, i.e., vectors, to the nodes of the input graph. These embeddings represent the raw concept of the node, without taking the graph structure into consideration. Hence it mainly reflects the syntactic meaning of the node. They can be learned or computed from the syntax. If we fix a signature, we could learn a basic embeddings for all sort and operator symbols of the signature, assign variables a non-learnable embedding computed just from the name, and learn some embedding merging operations to generate embeddings of more complex expressions or contexts. These merging operations can be performed by small encoder-only transformer models for example. In the case where we have no fixed signature, i.e., we want to create a foundational model to be used with any given signature we could just learn the aggregation operations, but this is left for further research.

GNN Body: The exact architecture of the message passing GNN has to be determined, but its purpose is to enrich the basic embeddings into structure-aware embeddings. It is important that the architecture selected expects directed edges, as the direction of morphisms is key in this setup.

Conjecturer: Just a linear projection layer from the original embedding dimension to a 3-dimensional vector. This means that the conjecturer will be classifying nodes into 3 categories: OBJECTIVE, INCLUDE and EXCLUDE, with the constraint that only one node can be chosen to be the objective node, so first we take the node with the highest OBJECTIVE score, mark it as an Objective, then we perform soft-max with the other two components of the other nodes to assign the labels INCLUDE and EXCLUDE. Then the challenge the Prover will try to solve is to generate a section of the OBJECTIVE node starting from the graph composed by the OBJECTIVE node and all the nodes labeled as INCLUDE. Morphisms that exist due to composition or currying of morphisms that include an excluded node won't be included.

Prover: this component is just a linear layer from the original embedding dimension to a 3-dimensional vector, same as the conjecturer, and then apply a soft-max. It classifies the nodes in 3 categories: KEEP, EXPAND and DELETE.

Now we have to consider training the model. The first problem is cold start. At the beginning we face an empty graph to feed to the conjecturer and a conjecturer yielding nonsense. This means that we have to somehow pre-train the conjecturer before stating the unsupervised training loop. We will propose a way to go around this problem, but it must undergo testing before assuring it works. We start by feeding the model a type we know it will find a type in relatively few steps of expansion, like $A \rightarrow A$ in our previous example. We can use this expansion to construct a base graph. This graph contains objects whose sections are fairly easy to construct, as we did it in a few iterations, then we select some of them and use them first as labeled examples for our conjecturer. Then we start the unsupervised training loop. At first, our prover also yields nonsense, so if we let it decide the path of our partial expansion it will fail catastrophically, leaving us with no examples to train our model with. To solve this, we consider a function that determines the score (value) needed by a label in order for us to take action. Name this function $f: \omega \rightarrow (0, +\infty)$. As an example, we take the function $f = 0.8$, then if we get the output $(0.3, 0.3, 0.4)$ from the prover, that node would still be label EXPAND, although it should be labeled as DELETE. In the other hand, if the prover output $(0.05, 0.1, 0.85)$ for a node, it would be labeled as DELETE. A constant function provides little-to-none benefit, because it will cap the complexity of objects our model can handle, as by not allowing drastic pruning the model won't be able to dig to deep into the graph. An example of such a function could be $f(x) = (a - b) \exp^{-cx} + b$, with $a, b, c > 0$. This confidence baseline $f(x)$ will decay over time, when x grows, and it will give more autonomy to the Prover as it learns to accurately slice through the combinatorial explosion.

Chapter 4

Conclusion

Appendix A

Derivation Rules

We denote "from $S_1, \dots, S_n$ derive S ", where $S_1, \dots, S_n, S$ are rules or sequents, as follows:

$$\frac{S_1 \quad \dots \quad S_n}{S}$$

Also, we write

$$\frac{S_1}{S_2} \text{ abbreviates } \frac{S_1}{S_2} \text{ and } \frac{S_2}{S_1}$$

A.1 Derivation Rules of Generalized Algebraic Theories

In this section we present the derivation rules of Generalized Algebraic Theories as presented in [4], but with a more compact notation.

$$\text{LI1 } \frac{\Gamma \vdash A \text{ type}}{\Gamma \vdash A = A}$$

$$\text{LI2 } \frac{\Gamma \vdash t : A}{\Gamma \vdash t = t : A}$$

$$\text{LI3 } \frac{\Gamma \vdash A_1 = A_2}{\Gamma \vdash A_2 = A_1}$$

$$\text{LI4 } \frac{\Gamma \vdash t_1 = t_2 : A}{\Gamma \vdash t_2 = t_1 : A}$$

$$\text{LI5 } \frac{\Gamma \vdash A_1 = A_2 \quad \Gamma \vdash A_2 = A_3}{A_1 = A_3}$$

$$\text{LI6 } \frac{\Gamma \vdash t_1 = t_2 : A \quad \Gamma \vdash t_2 = t_3 : A}{\Gamma \vdash t_1 = t_3 : A}$$

$$\text{LI7 } \frac{\Gamma \vdash t_1 = t_2 : A_1 \quad \Gamma \vdash A_1 = A_2}{t_1 = t_2 : A_2}$$

$$\text{T1} \quad \frac{\Gamma \vdash A_1 = A_2 \quad \Gamma \vdash t : A_1}{\Gamma \vdash t : A_2}$$

CF1 For $n \geq 0, 1 \leq i \leq n+1$:

$$\frac{x_1 : A_1, \dots, x_n : A_n \vdash A_{n+1} \text{ type}}{x_1 : A_1, \dots, x_{n+1} : A_{n+1} \vdash x_i : A_i}$$

providing that x_{n+1} is a variable distinct from all $x_1, \dots, x_n$.

CF2(A) For every sort symbol S with well-formed introductory rule

$$x_1 : A_1, \dots, x_n : A_n \vdash S(x_1, \dots, x_n) \text{ type},$$

for every context Γ ,

$$\frac{\Gamma \vdash t_1 : A_1 \quad \Gamma \vdash t_2 : A_2[x_1 \mid t_1] \dots \Gamma \vdash t_n : A_n[x_1 \mid t_1, \dots, x_{n-1} \mid t_{n-1}]}{S(t_1, \dots, t_n) \text{ type}}$$

CF2(B) For every operator symbol F with well-formed introductory rule

$$x_1 : A_1, \dots, x_n : A_n \vdash F(x_1, \dots, x_n) : A,$$

for every context P ,

$$\frac{\Gamma \vdash t_1 : A_1 \quad \Gamma \vdash t_2 : A_2[x_1 \mid t_1] \dots \Gamma \vdash t_n : A_n[x_1 \mid t_1, \dots, x_{n-1} \mid t_{n-1}]}{\Gamma \vdash F(t_1, \dots, t_n) : A[x_1 \mid t_1, \dots, x_n \mid t_n]}$$

SI1 If Γ is a context, then

$$\frac{x_1 : A_1, \dots, x_n : A_n \vdash A = A' \quad \Gamma \vdash s_1 = s'_1 : A_1 \quad \Gamma \vdash s_2 = s'_2 : A_2[x_1 \mid s_1] \dots \Gamma \vdash s_m = s'_m : A_n[x_1 \mid s_1, \dots, x_n \mid s_n]}{\Gamma \vdash A[x_1 \mid s_1, \dots, x_n \mid s_n] = A'[x_1 \mid s'_1, \dots, x_n \mid s'_n]}$$

SI2 If Γ is a context, then

$$\frac{x_1 : A_1, \dots, x_n : A_n \vdash s = s' : A \quad \Gamma \vdash s_1 = s'_1 : A_1 \quad \Gamma \vdash s_2 = s'_2 : A_2[x_1 \mid s_1] \dots \Gamma \vdash s_n = s'_n : A_n[x_1 \mid s_1, \dots, x_n \mid s_n]}{s[x_1 \mid s_1, \dots, x_n \mid s_n] = s'[x_1 \mid s'_1, \dots, x_n \mid s'_n]}$$

A1 If $x_1 : A_1, \dots, x_n : A_n \vdash A = A'$ is an axiom, then

$$\frac{x_1 : A_1, \dots, x_n : A_n \vdash A \text{ type} \quad x_1 : A_1, \dots, x_n : A_n \vdash A' \text{ type}}{x_1 : A_1, \dots, x_n : A_n \vdash A = A'}$$

A2 If $x_1 : A_1, \dots, x_n : A_n \vdash t = t' : A$ is an axiom, then from

$$\frac{x_1 : A_1, \dots, x_n : A_n \vdash t : A \quad x_1 : A_1, \dots, x_n : A_n \vdash t' : A}{x_1 : A_1, \dots, x_n : A_n \vdash t = t' : A}$$

A.2 Derivation Rules of Calculus of Constructions

In this section we present all the derivation rules for CoC as presented in [11].

Context Formation Rules:

$$\begin{array}{l}
 \text{EMPTY} \quad \frac{}{\text{ok}} \\
 \text{CONT-INTRO} \quad \frac{\Gamma \vdash A \text{ type}}{\Gamma, x: A \text{ ok}}, \text{ if } x \notin \text{Var}(\Gamma) \\
 \text{CONT-THIN} \quad \frac{\Gamma, \Delta \text{ ok} \quad \Gamma \vdash A \text{ type}}{\Gamma, x: A, \Delta \text{ ok}}, \text{ if } x \notin \text{Var}(\Gamma) \cup \text{Var}(\Delta) \\
 \text{CONT-SUB} \quad \frac{\Gamma, x: A, \Delta \text{ ok} \quad \Gamma \vdash t: A}{\Gamma, \Delta[x \mid t] \text{ ok}}
 \end{array}$$

Context Equality Rules

$$\begin{array}{l}
 \text{CREFL} \quad \frac{\Gamma = \Gamma}{\Gamma \text{ ok}} \\
 \text{CSYM} \quad \frac{\Gamma = \Delta}{\Delta = \Gamma} \\
 \text{CTRANS} \quad \frac{\Gamma = \Delta \quad \Delta = \Theta}{\Gamma = \Theta} \\
 \text{CEQU} \quad \frac{\Gamma, x: A, \Delta \text{ ok} \quad \Gamma \vdash A = B}{\Gamma, x: A, \Delta = \Gamma, x: B, \Delta} \\
 \text{CREFLECT1} \quad \frac{\Gamma, x: A = \Delta, x: B}{\Gamma = \Delta} \\
 \text{CREFLECT2} \quad \frac{\Gamma, x: A = \Delta, x: B}{\Gamma \vdash A = B_1}
 \end{array}$$

Structural Rules for Judgement Formation

$$\begin{array}{l}
 \text{ASS} \quad \frac{\Gamma, x: A, \Delta \text{ ok}}{\Gamma, x: A, \Delta \vdash x: A} \\
 \text{REFLECTION1} \quad \frac{\Gamma \vdash \mathcal{J}}{\Gamma \text{ ok}} \\
 \text{REFLECTION2} \quad \frac{\Gamma \vdash t: A}{\Gamma \vdash A \text{ type}} \\
 \text{THIN} \quad \frac{\Gamma, \Delta \vdash \mathcal{J} \quad \Gamma \vdash A \text{ type}}{\Gamma, x: A, \Delta \vdash \mathcal{J}}, \text{ if } x \notin \text{Var}(\Gamma) \cup \text{Var}(\Delta) \\
 \text{SUB} \quad \frac{\Gamma, x: A, \Delta \vdash \mathcal{J} \quad \Gamma \vdash t: A}{\Gamma, \Delta[x \mid t] \vdash \mathcal{J}[x \mid t]}
 \end{array}$$

Equality Rules:

$$\begin{array}{l}
\text{EQU} \quad \frac{\Gamma = \Delta \quad \Gamma \vdash \mathcal{J}}{\Delta \vdash \mathcal{J}} \\
\text{REFL-TYP} \quad \frac{\Gamma \vdash A = A}{\Gamma \vdash A \text{ type}} \\
\text{REFL-OBJ} \quad \frac{\Gamma \vdash t = t : A}{\Gamma \vdash t : A} \\
\text{SYMM-TYP} \quad \frac{\Gamma \vdash A = B}{\Gamma \vdash B = A} \\
\text{SYMM-OBJ} \quad \frac{\Gamma \vdash t = s : A}{\Gamma \vdash s = t : A} \\
\text{TRANS-TYP} \quad \frac{\Gamma \vdash A = B \quad \Gamma \vdash B = C}{\Gamma \vdash A = C} \\
\text{TRANS-OBJ} \quad \frac{\Gamma \vdash t_1 = t_2 : A \quad \Gamma \vdash t_2 = t_3 : A}{\Gamma \vdash t_1 = t_3 : A} \\
\text{REPL1} \quad \frac{\Gamma \vdash t = s : A \quad \Gamma, x : A, \Delta \text{ ok}}{\Gamma, \Delta[x \mid t] = \Gamma, \Delta[x \mid t]} \\
\text{REPL2} \quad \frac{\Gamma \vdash t = s : A \quad \Gamma, x : A, \Gamma \vdash B \text{ type}}{\Gamma, \Delta[x \mid t] \vdash B[x \mid t] = B[x \mid s]} \\
\text{REPL3} \quad \frac{\Gamma \vdash t_1 = t_2 : A \quad \Gamma, x : A, \Delta \vdash s : B}{\Gamma, \Delta[x \mid t_1] \vdash s[x \mid t_1] = s[x \mid t_2] : B[x \mid t_1]} \\
\text{CONV1} \quad \frac{\Gamma \vdash A = B \quad \Gamma \vdash t : A}{\Gamma \vdash t : B} \\
\text{CONV2} \quad \frac{\Gamma \vdash A = B \quad \Gamma \vdash t = s : A}{\Gamma \vdash t = s : B}
\end{array}$$

Rules for Products of types

$$\begin{array}{l}
\Pi\text{-FORM} \quad \frac{\Gamma, x : A \vdash B \text{ type}}{\Gamma \vdash (\Pi x : A) B \text{ type}} \\
\Pi\text{-EQU} \quad \frac{\Gamma \vdash A_1 = A_2 \quad \Gamma, x : A_1 \vdash B_1 = B_2}{\Gamma \vdash (\Pi x : A_1) B_1 = (\Pi x : A_2) B_2} \\
\Pi\text{-INTRO} \quad \frac{\Gamma, x : A \vdash t : B}{\Gamma \vdash (\lambda x : A) t : (\Pi x : A) B} \\
\zeta\text{-RULE} \quad \frac{\Gamma, x : A_1 \vdash t_1 = t_2 : B \quad \Gamma \vdash A_1 = A_2}{\Gamma \vdash (\lambda x : A_1) t_1 = (\lambda x : A_2) t_2 : (\Pi x : A_1) B}
\end{array}$$

$$\Pi\text{-ELIM} \frac{\Gamma \vdash t : (\prod x : A)B \quad \Gamma \vdash s : A \quad \Gamma, x : A \vdash B \text{ type}}{\Gamma \vdash \text{App}([x : A]B, t, s) : B[x \mid s]}$$

$$\Pi\text{-ELIM-EQU} \frac{\Gamma \vdash A_1 = A_2 \quad \Gamma \vdash t_1 = t_2 : (\prod x : A_1)B_1 \quad \Gamma, x : A_1 \vdash B_1 = B_2 \quad \Gamma \vdash s_1 = s_2 : A_1}{\Gamma \vdash \text{App}([x : A_1]B_1, t_1, s_1) = \text{App}([x : A_2]B_2, t_2, s_2) : B_1[x \mid s_1]}$$

$$\beta\text{-RULE} \frac{\Gamma, x : A \vdash B \text{ type} \quad \Gamma, x : A \vdash t : B \quad \Gamma \vdash s : A}{\Gamma \vdash \text{App}([x : A]B, (\lambda x : A)t, s) = t[x \mid s] : B[x \mid s]}$$

$$\eta\text{-RULE} \frac{\Gamma, x : A \vdash B \text{ type} \quad \Gamma \vdash t : (\prod x : A)B}{\Gamma \vdash (\lambda x : A)\text{App}([x : A]B, t, x) = t : (\prod x : A)B}$$

Rules for Propositions

$$\text{PROP} \frac{\Gamma \text{ ok}}{\Gamma \vdash \text{Prop type}}$$

$$\text{PROP-TYPE} \frac{\Gamma \vdash p : \text{Prop}}{\Gamma \vdash \text{Proof}(p) \text{ type}}$$

$$\text{PROP-EQU} \frac{\Gamma \vdash p_1 = p_2 : \text{Prop}}{\Gamma \vdash \text{Proof}(p_1) = \text{Proof}(p_2)}$$

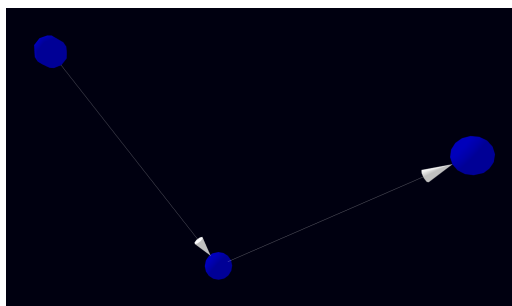
$$\forall\text{-INTRO} \frac{\Gamma, x : A \vdash p : \text{Prop}}{\Gamma \vdash (\forall x : A)p : \text{Prop}}$$

$$\forall\text{-EQU} \frac{\Gamma, x : A_1 \vdash p_1 = p_2 : \text{Prop} \quad \Gamma \vdash A_1 = A_2}{\Gamma \vdash (\forall x : A_1)p_1 = (\forall x : A_2)p_2 : \text{Prop}}$$

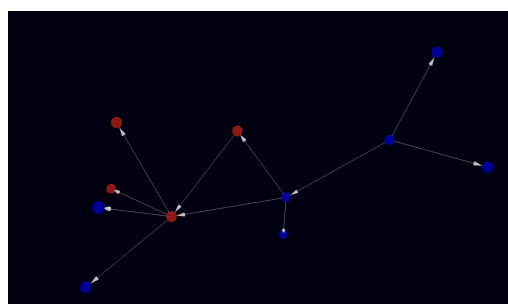
$$\forall\text{-ELIM} \frac{\Gamma, x : A \vdash p \in \text{Prop}}{\Gamma \vdash \text{Proof}((\forall x : A)p) = (\prod x : A)\text{Proof}(p)}$$

Appendix B

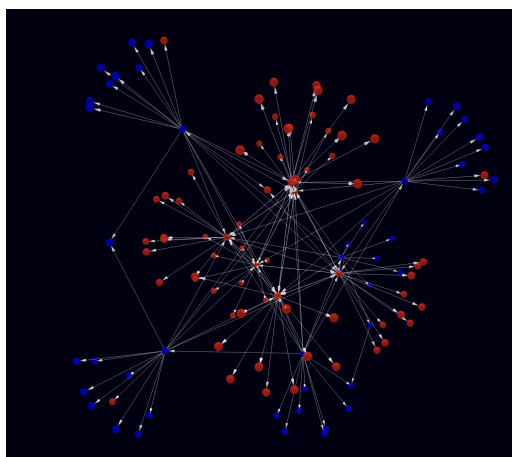
Figures



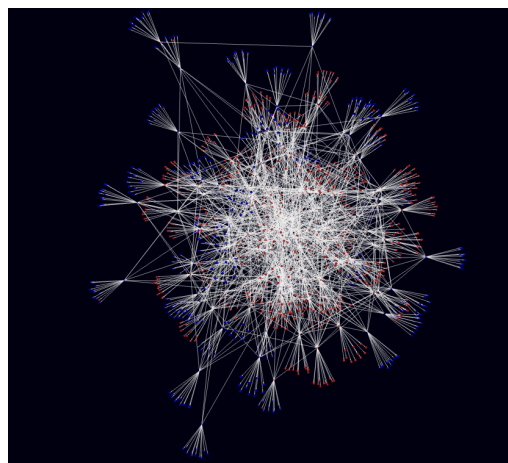
(a) The G_1 subcategory



(b) The G_2 subcategory



(c) The G_3 subcategory



(d) The G_4 subcategory

Figure B.1: Overview of the G_1 through G_4 subcategories.

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